ECSA 2025
LIMASSOL - CYPRUS

Software Architecture in the Age of Agentic AI

Karthik Vaidhyanathan and Henry Muccini

ECSA 2025 Tutorial Session, Sep 15 2025

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY
HYDERABAD

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

FRAMELAB

Welcome

SCAN ME

Part 1: Introductions

Getting to know each other and setting the agenda

Part 3: LLM for SA

Using LLM for Software Architecture Knowledge management and other tasks.

Part 2: SA and ML

How the field at the intersection of SA and ML have evolved over the years

Part 4: Agentic SA

Drive innovation and tech adoption. Successfully implement 2 major tech upgrades within the next yr that leads to 20% increase in cost savings.

About Us

Dr. Karthik Vaidhyanathan, PhD
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IIIT Hyderabad, India

Prof. Henry Muccini, PhD
FrAmeLab @SWEN
University of L'Aquila, Italy

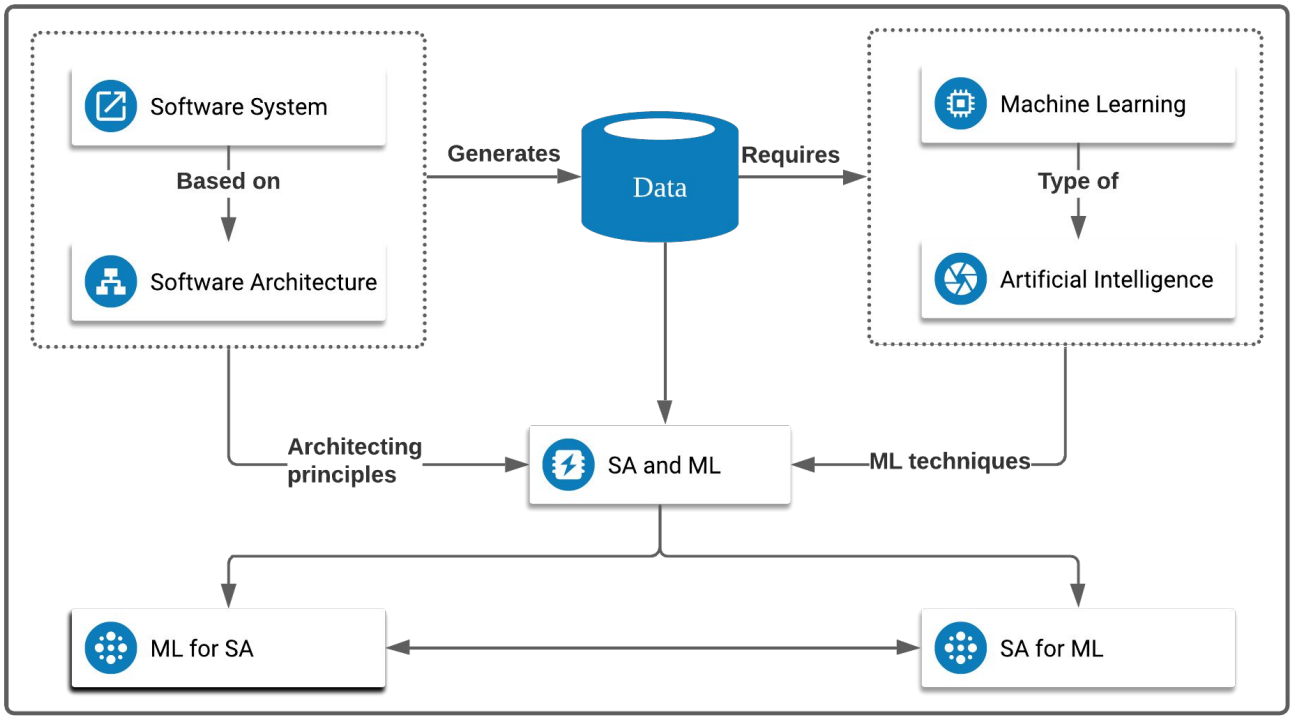
About you

Name/Surname

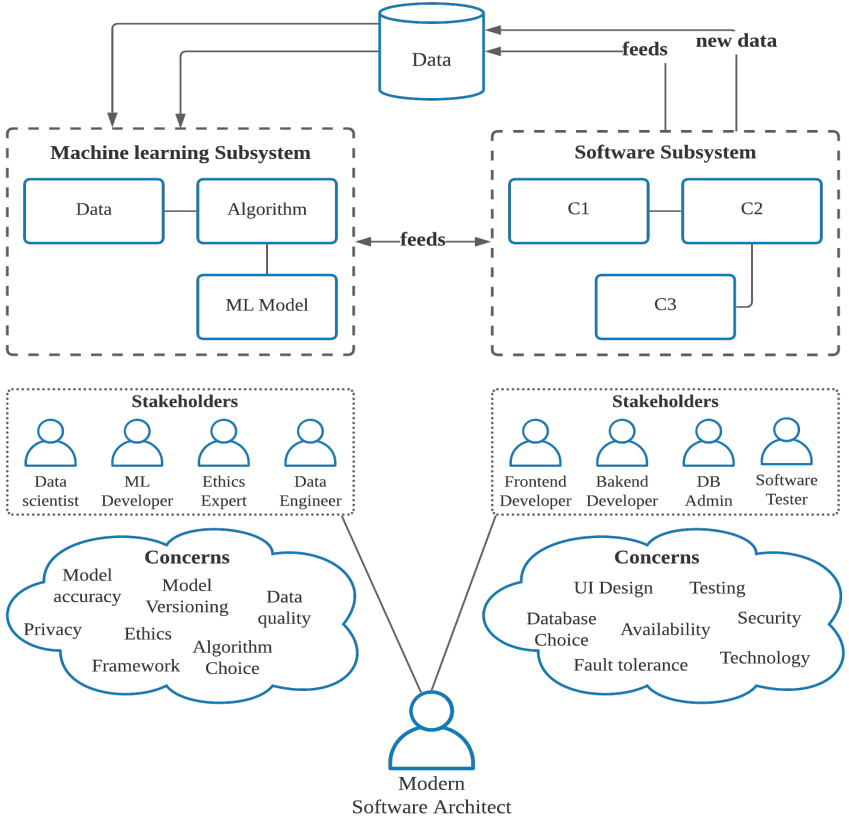
Skills on ML/SA

Interests on the topic

At the Intersection of Software Architecture and ML



High-level view of ML-enabled System



H. Muccini, K. Vaidhyanathan: **Software Architecture for ML-based Systems: What Exists and What Lies Ahead.** WAIN@ICSE 2021. Image: courtesy of K. Vaidhyanathan

SA and ML: our journey so far (events)

2020:
ECSA'20: Theme on SA

2021:
ICSA'21: Working session on SA and ML

2023:
Dagstuhl on Software Architecture and Machine Learning, 2023

SEAA2025

2021:
JSS: Special Issue on "Software Architecture and Artificial Intelligence"

2022:
ICSA'22: Working session on SA and ML

2025:
ICSA'25: Working session on Architects Skills with GenAI

SA&ML – ICSA Workshop on SA and Machine Learning (2021-today)

CAIN – AI ENGINEERING CONFERENCE @ICSE (2022-today)

ML4SA
Using ML to improve SA design

ICSA_C 2019

A Machine Learning-driven Approach for Proactive Decision Making in Adaptive Architectures

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#self-adaptation

ICSA 2020

Quantitative Verification-Aided Machine Learning: A Tandem Approach for Architecting Self-Adaptive IoT Systems

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#verification

ACM/SAC 2024

ML-enabled Service Discovery for Microservice Architecture: A QoS Approach

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#microservice discovery

SA4ML / SA&ML

WAIN 2021

Software Architecture for ML-based Systems: What Exists and What Lies Ahead

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#SA of ML, #vision

IEEE Software 2022

#ML, #Agile, #Agile4ML

Agile4MLS—Leveraging Agile Practices for Developing Machine Learning-Enabled Systems

An Industrial Experience

Dagstuhl 2023

Report from Dagstuhl Seminar 23302

Software Architecture and Machine Learning

Grace A. Lewis[*1], Henry Muccini[*2], Ipek Ozkaya[3], Karthik Vaidhyanathan[†4], Roland Weiss[*5], and Liming Zhu[*6]

#SA & ML, #QualityAttributes

Some Concepts

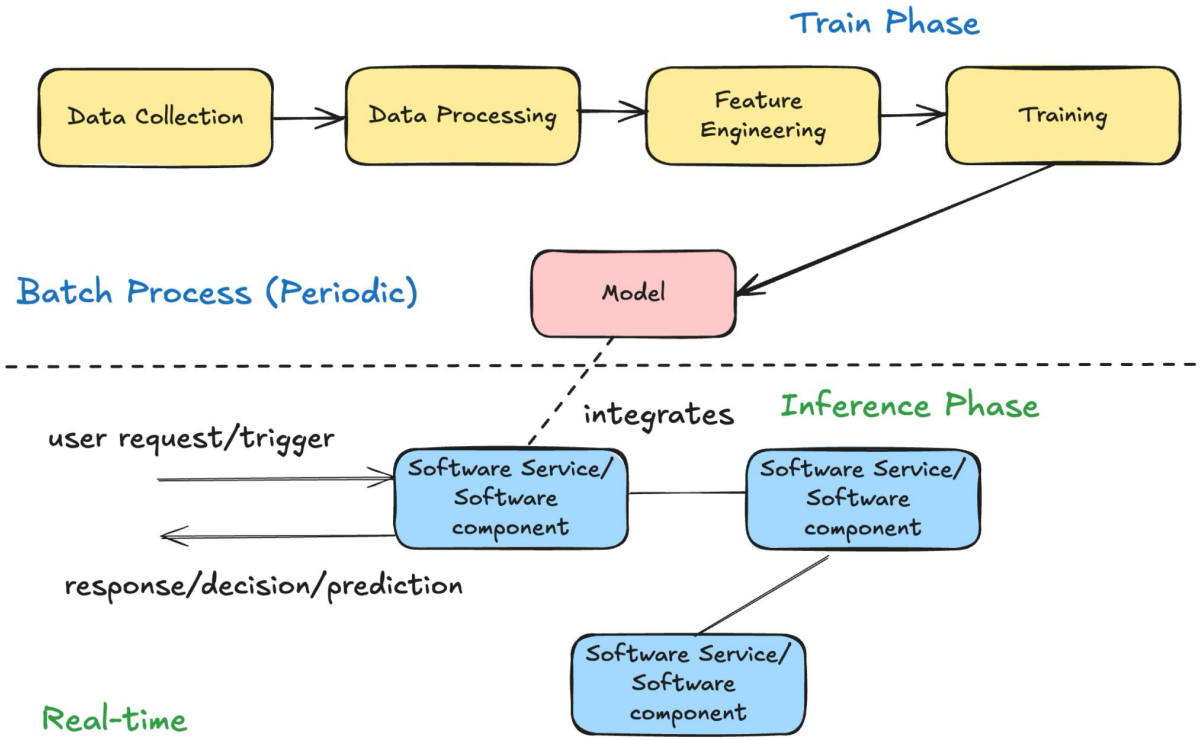
ML model

Dataset

Training

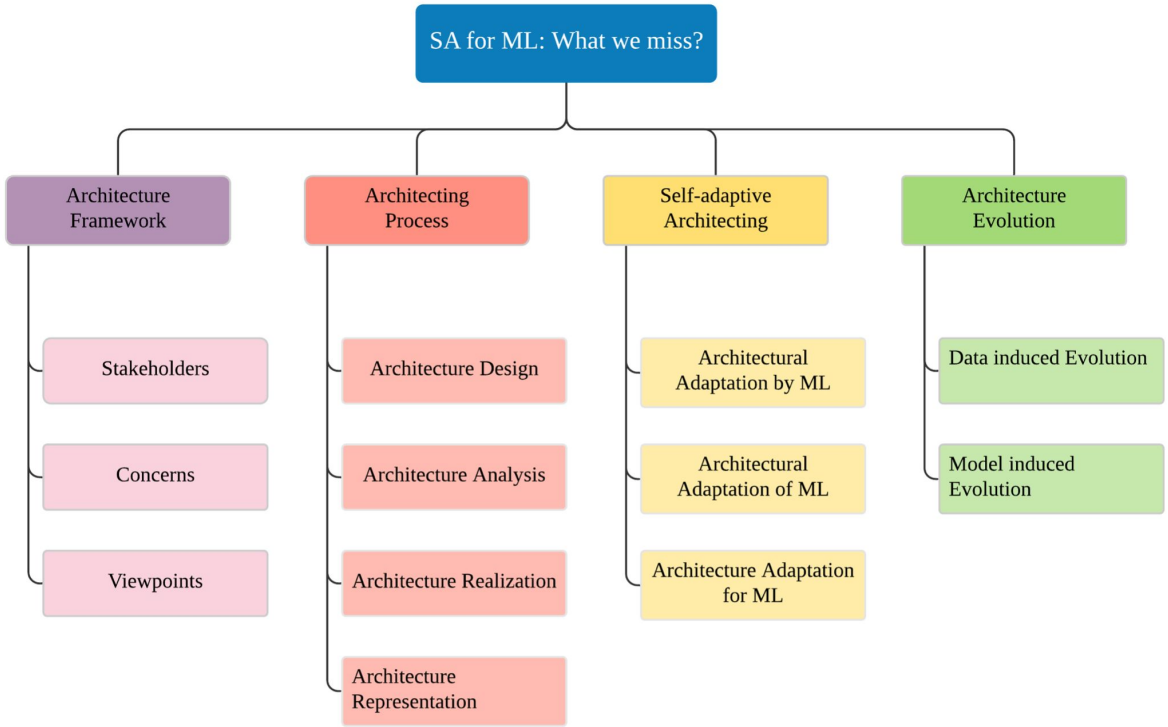
Testing

Evaluation Metrics



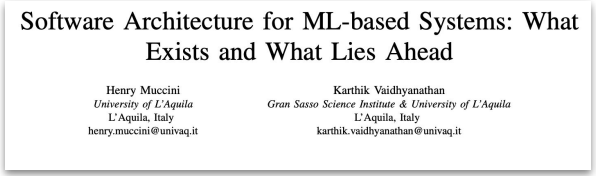
Part 2.1: SA for ML

SA for ML-enabled System: What we miss?



H. Muccini, K. Vaidhyanathan: **Software Architecture for ML-based Systems: What Exists and What Lies Ahead.** WAIN@ICSE 2021. Image: courtesy of K. Vaidhyanathan

Some of the works over the years in the area of SA4ML


Software Architecture for ML-based Systems: What Exists and What Lies Ahead

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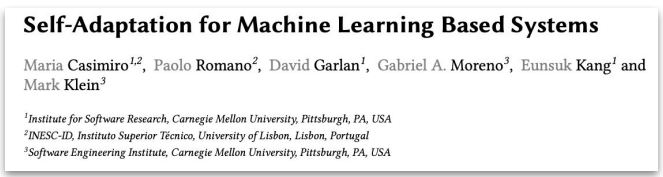
WAIN@ICSE2021


Self-Adaptation for Machine Learning Based Systems

Maria Casimiro[1,2], Paolo Romano[2], David Garlan[1], Gabriel A. Moreno[3], Eunsuk Kang[1] and Mark Klein[3]

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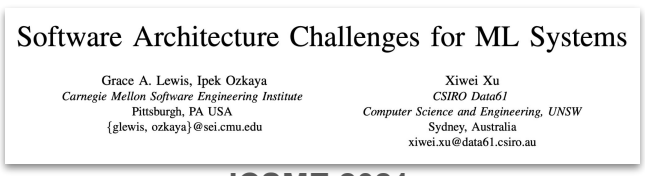
SAML@ECSA 2021


Software Architecture Challenges for ML Systems

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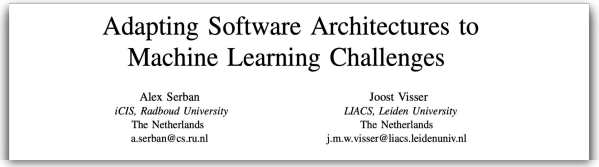
ICSME 2021


Adapting Software Architectures to Machine Learning Challenges

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SANER 2022

Architectural Design Decisions for Machine Learning Deployment

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ICSA 2022

Agile4MLS - Leveraging Agile Practices for Developing ML-enabled systems: An Industrial Experience

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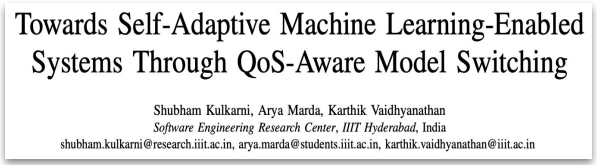
IEEE Software 2022


Towards Self-Adaptive Machine Learning-Enabled Systems Through QoS-Aware Model Switching

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ASE 2023

A Synthesis of Green Architectural Tactics for ML-Enabled Systems

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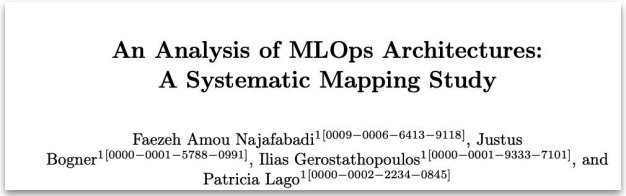
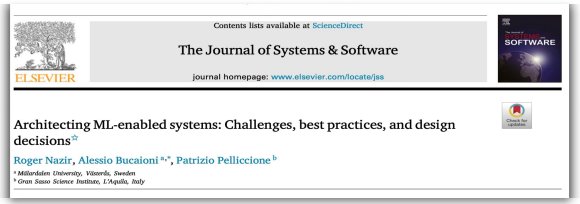
ICSE 2024


An Analysis of MLOps Architectures: A Systematic Mapping Study

Faezeh Amou Najafabadi[1 [0009−0006−6413−9118]], Justus Bogner[1 [0000−0001−5788−0991]], Ilias Gerostathopoulos[1 [0000−0001−9333−7101]], and Patricia Lago[1 [0000−0002−2234−0845]]

ECSA 2024


Contents lists available at ScienceDirect

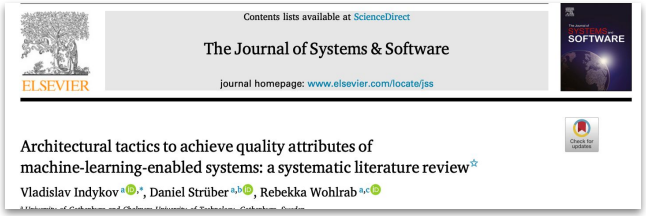
The Journal of Systems & Software

journal homepage: www.elsevier.com/locate/jss

Architecting ML-enabled systems: Challenges, best practices, and design decisions☆

Roger Nazir, Alessio Bucaioni, Patrizio Pelliccione

JSS 2024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The Journal of Systems & Software

journal homepage: www.elsevier.com/locate/jss

Architectural tactics to achieve quality attributes of machine-learning-enabled systems: a systematic literature review☆

Vladislav Indykov, Daniel Strüber, Rebekka Wohlrab

JSS 2025

HarmonE: A Self-Adaptive Approach to Architecting Sustainable MLOps

Hiya Bhatt[1⋆], Shaunak Biswas[1⋆], Srinivasan Rakhunathan[2], and Karthik Vaidhyanathan[1]

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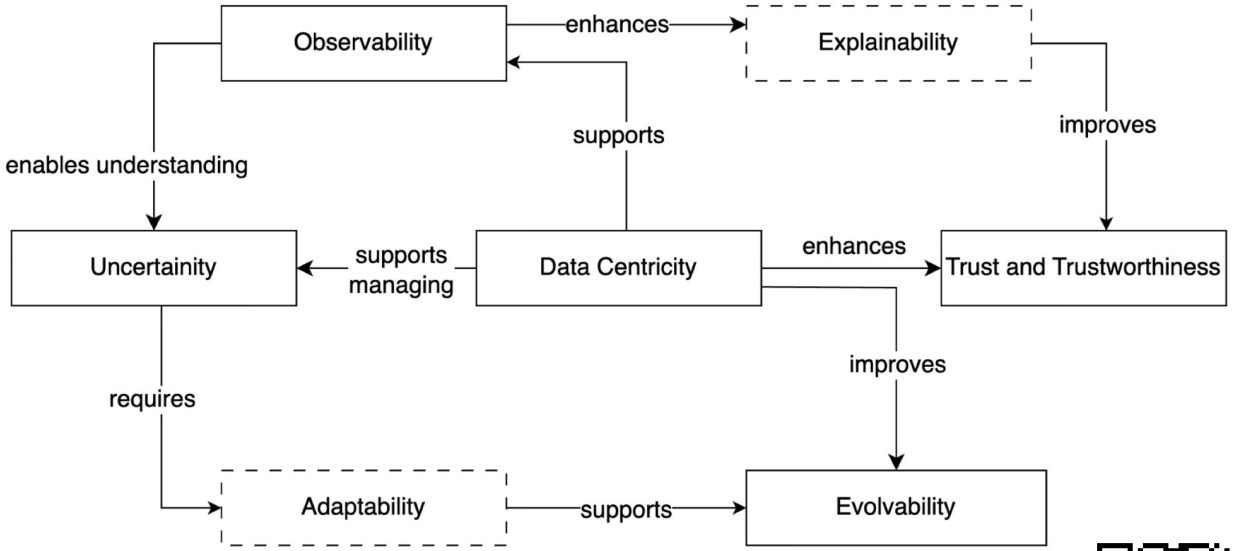
ECSA 2025

Quality-Centric View from our Dagstuhl Seminar



Report from Dagstuhl Seminar 23302

Software Architecture and Machine Learning

Grace A. Lewis*[1], Henry Muccini*[2], Ipek Ozkaya[3], Karthik Vaidhyanathan†[4], Roland Weiss*[5], and Liming Zhu*[6]

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Abstract

This report documents the program and outcomes of Dagstuhl Seminar 23302, "Software Architecture and Machine Learning". We summarize the goals and format of the seminar, results from the breakout groups, key definitions relevant to machine learning-enabled systems that were discussed, and the research roadmap that emerged from the discussions during the seminar. The report also includes the abstracts of the talks presented at the seminar and summaries of open discussions.

Seminar July 23–28, 2023 – https://www.dagstuhl.de/23302
2012 ACM Subject Classification Software and its engineering → Software architectures; Computing methodologies → Machine learning; Software and its engineering → Extra-functional properties; Computing methodologies → Artificial intelligence; Software and its engineering
Keywords and phrases Architecting ML-enabled Systems, ML for Software Architecture, Software Architecture for ML, Machine Learning, Software Architecture, Software Engineering
Digital Object Identifier 10.4230/DagRep.13.7.166

1 Executive Summary

Grace A. Lewis (Carnegie Mellon Software Engineering Institute – Pittsburgh, US)
Henry Muccini (University of L'Aquila, IT)
Ipek Ozkaya (Carnegie Mellon Software Engineering Institute – Pittsburgh, US)
Karthik Vaidhyanathan (IIIT – Hyderabad, IN)
Roland Weiss (ABB – Mannheim, DE)
Liming Zhu (Data61, CSIRO – Sydney, AU)

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© Grace A. Lewis, Henry Muccini, Ipek Ozkaya, Karthik Vaidhyanathan, Roland Weiss, and Liming Zhu

The pervasive and distributed nature of many of today's software systems requires making complex design decisions to guarantee important system qualities such as performance, reliability, safety and security. The practices within the field of software architecture guide the design and development of software systems from its high-level blueprint down to their implementation and operations. While the fundamentals of software architecture practices

* Editor / Organizer
† Editorial Assistant / Collector

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Software Architecture and Machine Learning, *Dagstuhl Reports*, Vol. 13, Issue 7, pp. 166–188
Editors: Grace A. Lewis, Henry Muccini, Ipek Ozkaya, Karthik Vaidhyanathan, Roland Weiss, and Liming Zhu
Dagstuhl Reports
Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl Publishing, Germany

Where we are and what we still miss?

Architecture Framework

Lack of generic framework for ML-enabled Systems. Some emerging phase wise frameworks

Architecting Process

Design and realization has been gaining some attention. While more works needs to be done on analysis and representation

Self-adaptive Architecting

Self-adaptation by ML has attracted lot of works. Self-adaptation of and for ML requires lot more attention

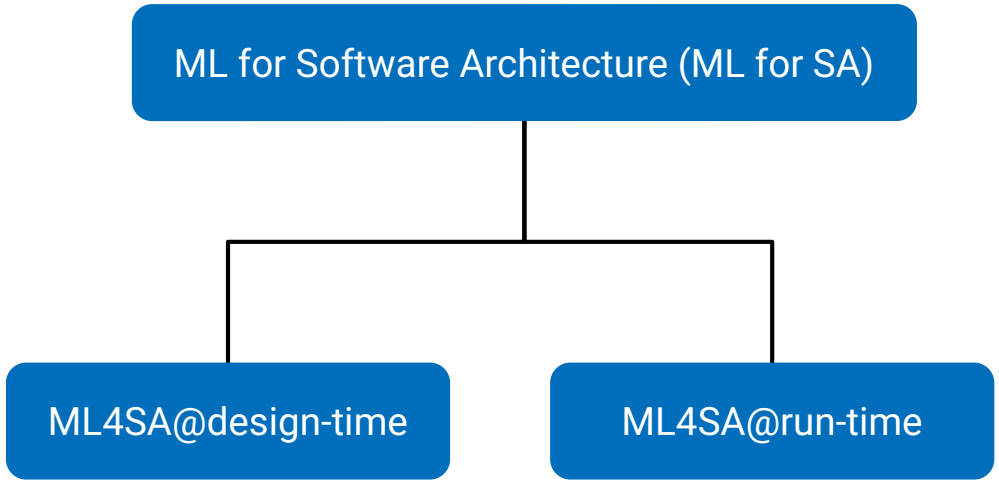
Architecture Evolution

While some works have emerged in the space of data and model induced evolution, more work on the long term impact and maintainability is required

What more? Time for Discussion

Part 2.2: ML for SA

What about ML for SA



Artificial Intelligence for Software Architecture: Literature Review and the Road Ahead

ALESSIO BUCAIONI, Mälardalen Universty, Sweden
MARTIN WEYSSOW, JUNDA HE, YUNBO LYU, and DAVID LO, Singapore Management University, Singapore

Limited focus in the pre-LLM era of using ML4SA. Most focus has been for ML4SA@run-time

ML4SA@design-time (Some early works)

Detecting, Tracing, and Monitoring Architectural Tactics in Code

Mehdi Mirakhorli, *Member, IEEE* and Jane Cleland-Huang, *Member, IEEE*

IEEE TSE 2015

Automatic Extraction of Design Decisions from Issue Management Systems: A Machine Learning Based Approach

Manoj Bhat*, Klym Shumaiev*, Andreas Biesdorf+, Uwe Hohenstein+, and Florian Matthes*

ECSA 2017

Architecture Anti-patterns: Automatically Detectable Violations of Design Principles

Ran Mo, Yuanfang Cai, Rick Kazman, Lu Xiao, and Qiong Feng

IEEE TSE 2019

Article

Towards Predicting Architectural Design Patterns: A Machine Learning Approach

Sirojiddin Komolov *,†, Gcinizwe Dlamini †, Swati Megha † and Manuel Mazzara †

Institute of Software Development and Engineering, Innopolis University, 420500 Innopolis, Russia

Computers 2022

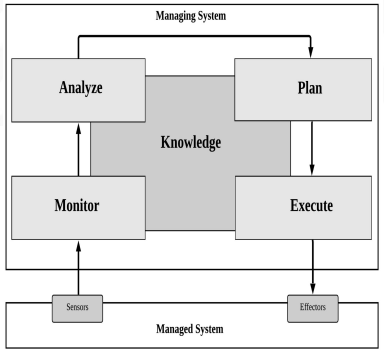
A Systematic Literature Review of Machine Learning Approaches for Migrating Monolithic Systems to Microservices

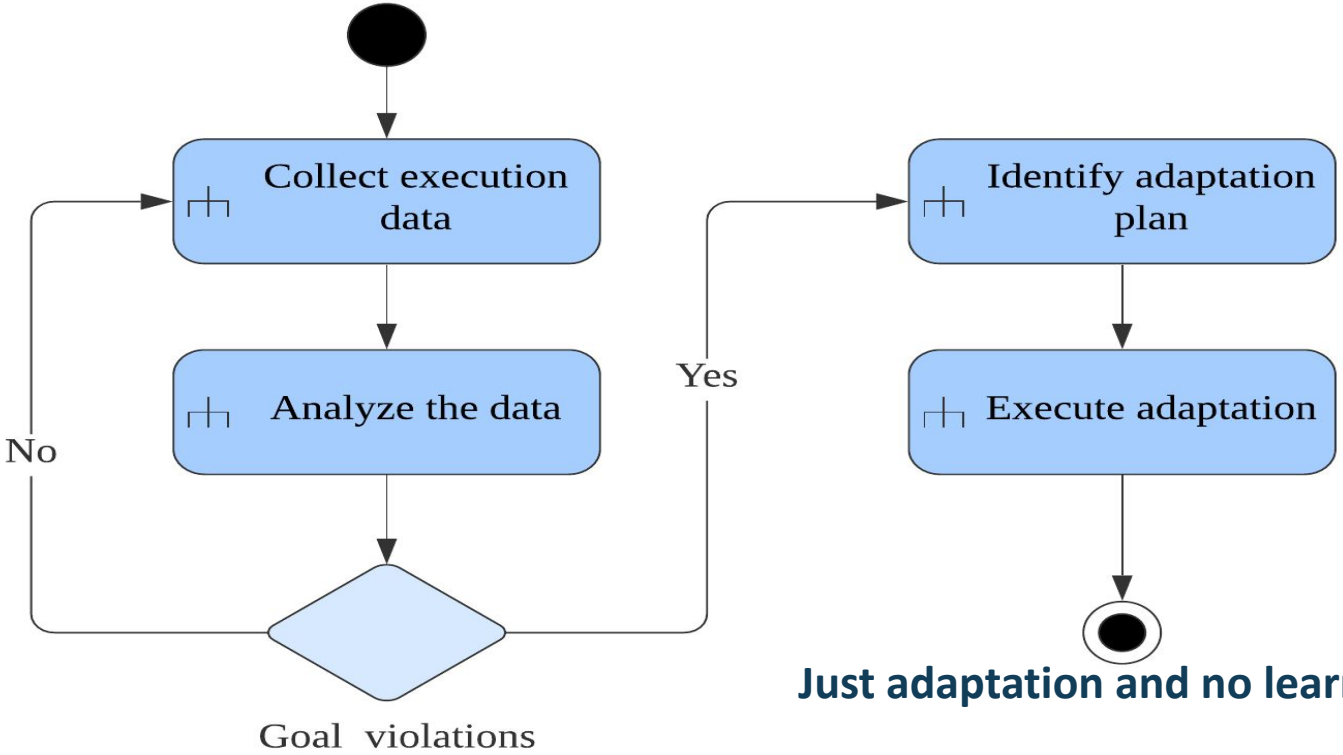
Imen Trabelsi, Brahim Mahmoudi, Jean Baptiste Minani, Naouel Moha, and Yann-Gaël Guéhéneuc

IEEE TSE 2025

Run-Time

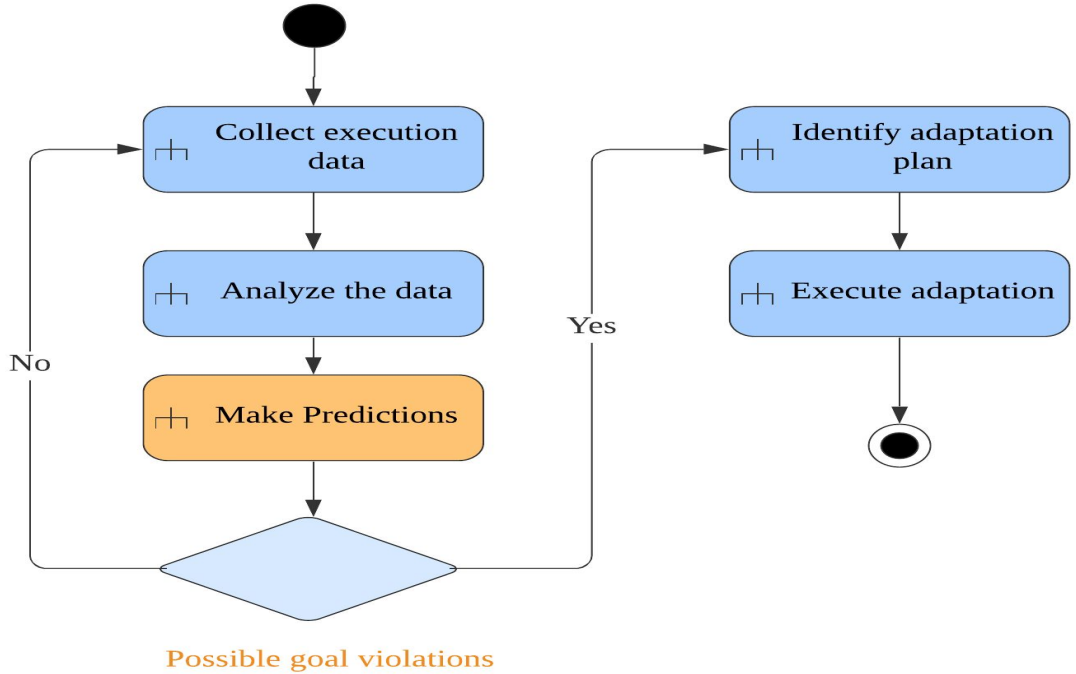
Self-adaptation traditional process - Reactive





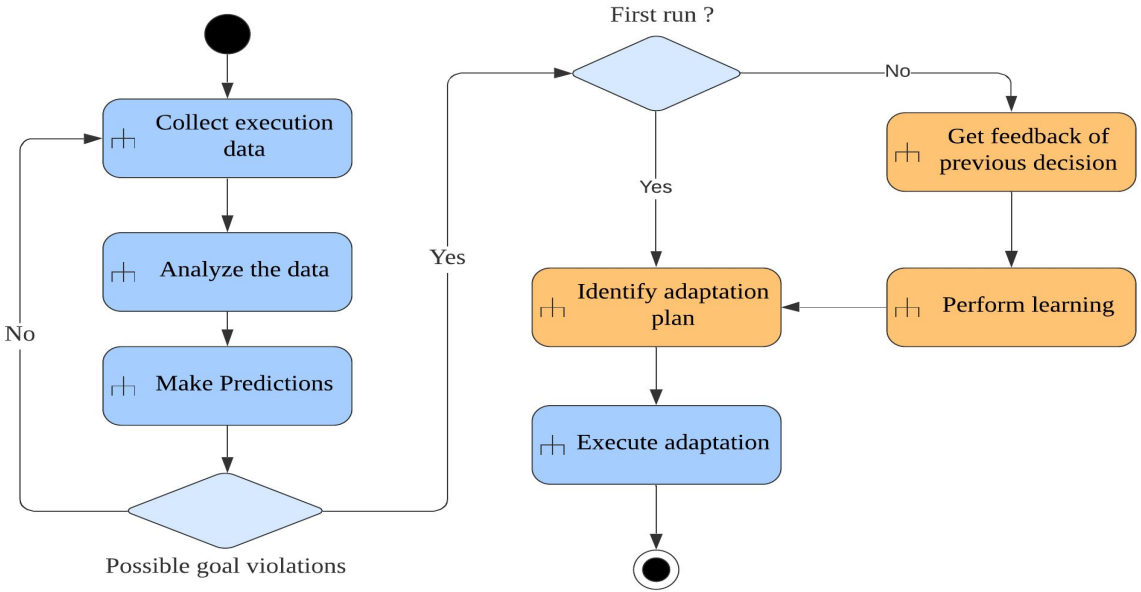
Just adaptation and no learning !!

Self-adaptation process using ML - Proactive



Henry Muccini and Karthik Vaidhyanathan. **A machine learning-driven approach for proactive decision making in adaptive architectures**. In 2019 IEEE International Conference on Software Architecture Companion (ICSA-C), pages 242-245. IEEE, 2019.

Self-adaptation process for learning - ML-driven

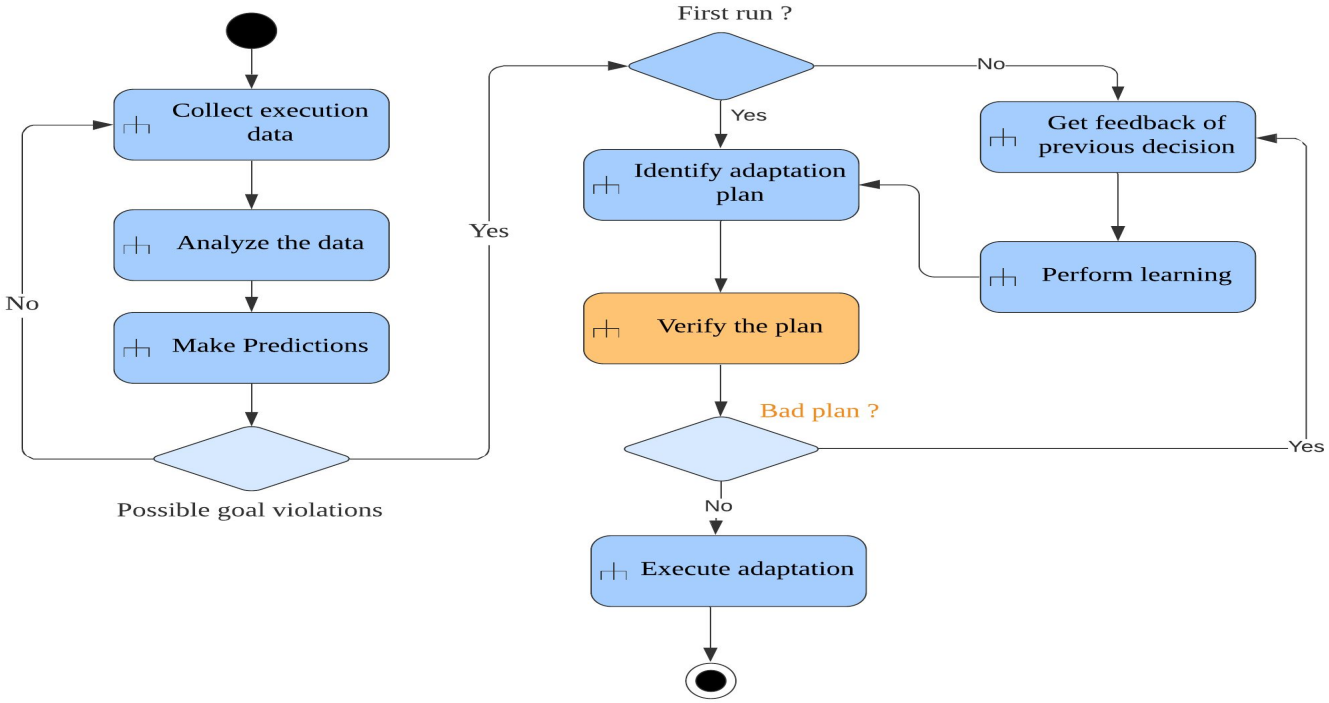


Using ML & RL

QoS: Energy efficiency and performance

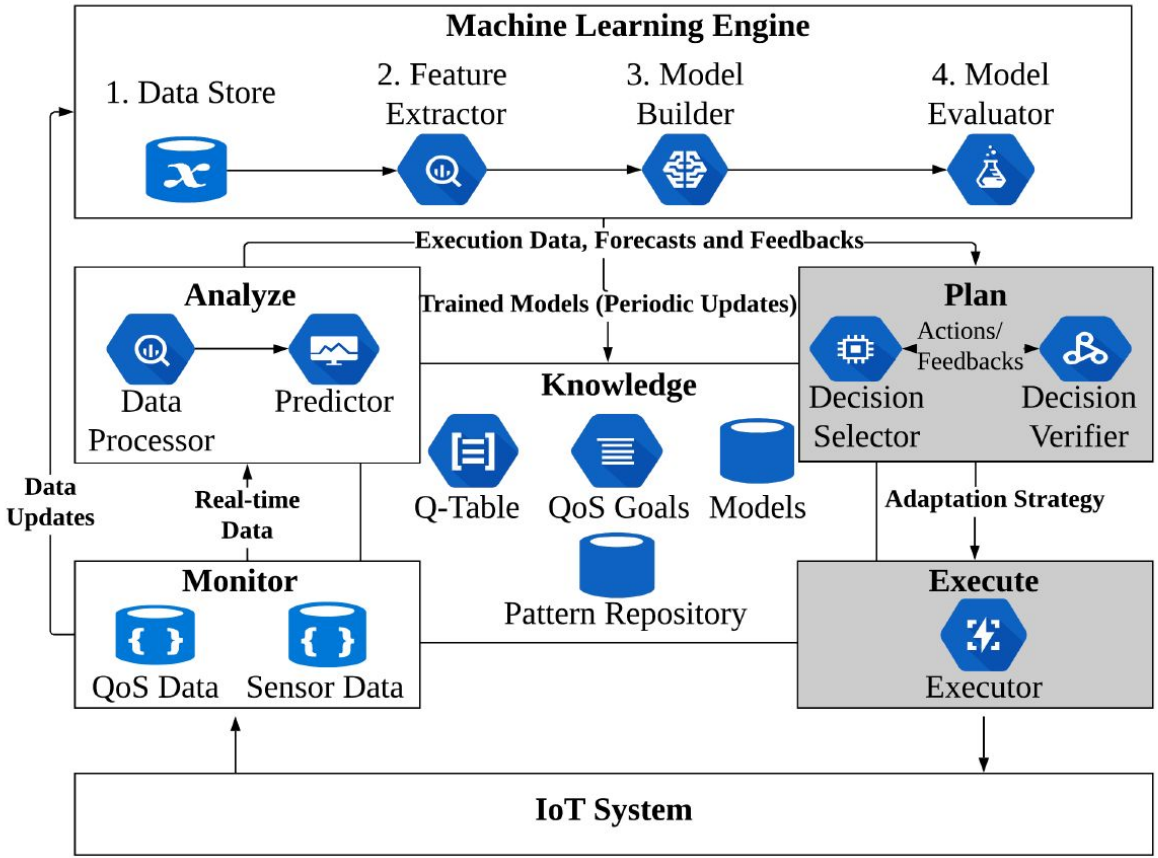
Henry Muccini, Karthik Vaidhyanathan. **Leveraging Machine Learning Techniques for Architecting Self-Adaptive IoT Systems**. Proceedings k ooo ooo b l m o z no o c o of the 6th IEEE International Conference on Smart Computing (SMARTCOMP 2020)

Self-adaptation process - Verification aided ML

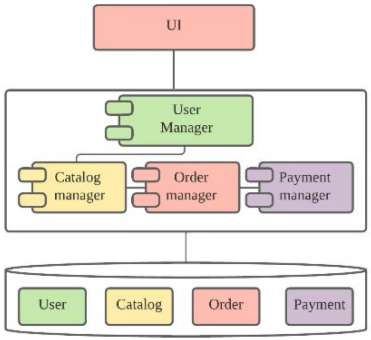


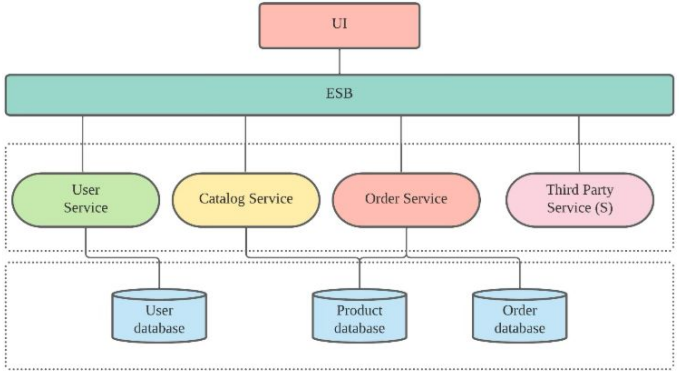
Javier Cámara Moreno, Henry Muccini, Karthik Vaidhyanathan. **Quantitative Verification-Aided Machine Learning: A Mixed-Method Approach for Architecting Self-Adaptive IoT Systems**. Proceedings of ICSA 2020

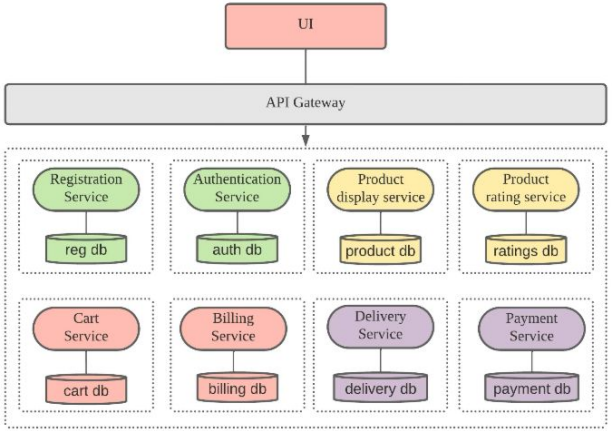
Approach overview



Evolution of Software Systems







1990 **2000** **2010** **2020**

Evolution of ML over the years

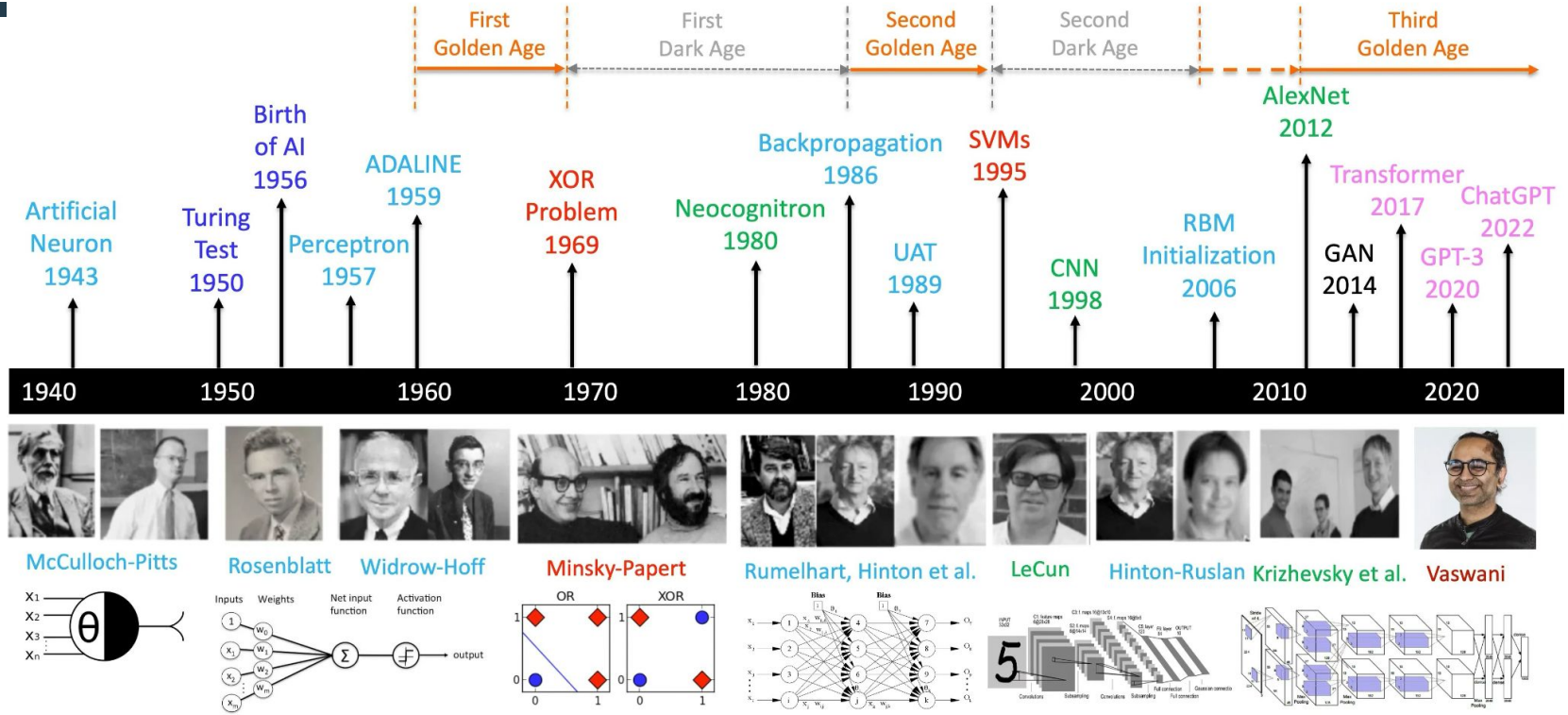
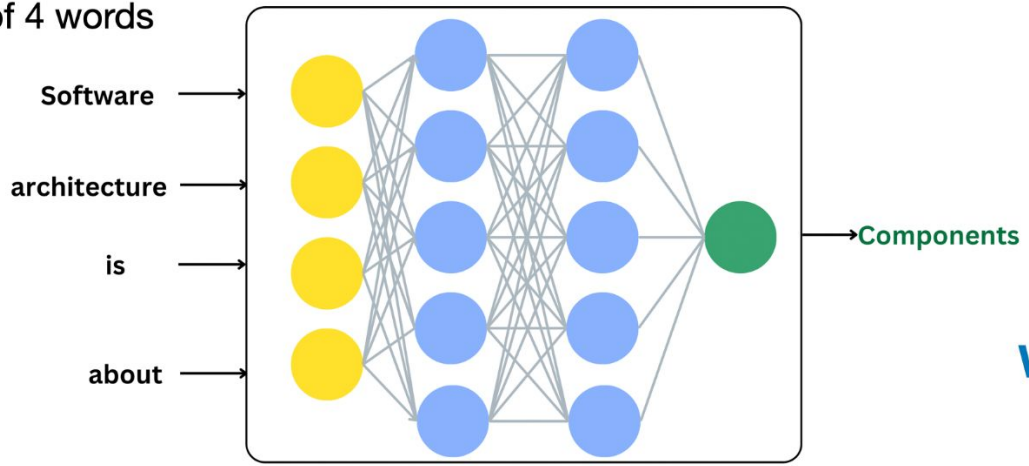


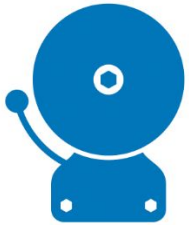
Image source: https://medium.com/@lmpo/a-brief-history-of-ai-with-deep-learning-26f7948bc87b

Part 3: LLM for SA

Journey to LLMs: Next Word Prediction Problem

Eg: Context of 4 words





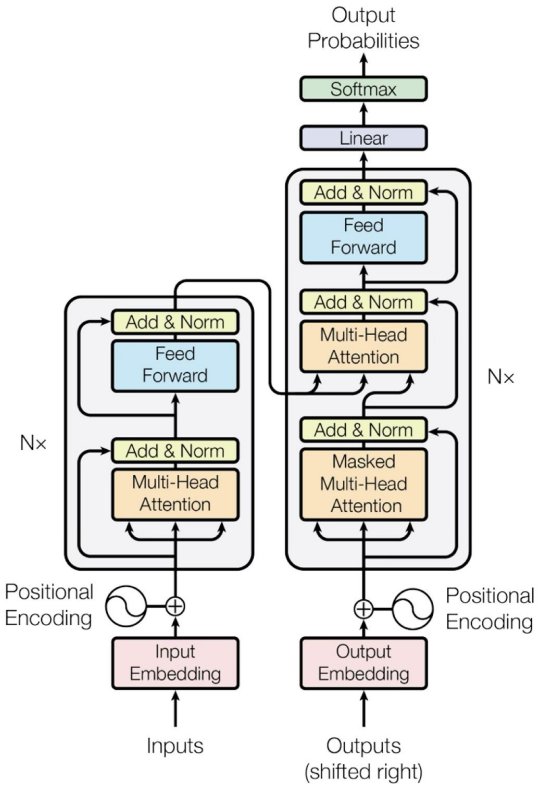
What does “it” refer to?

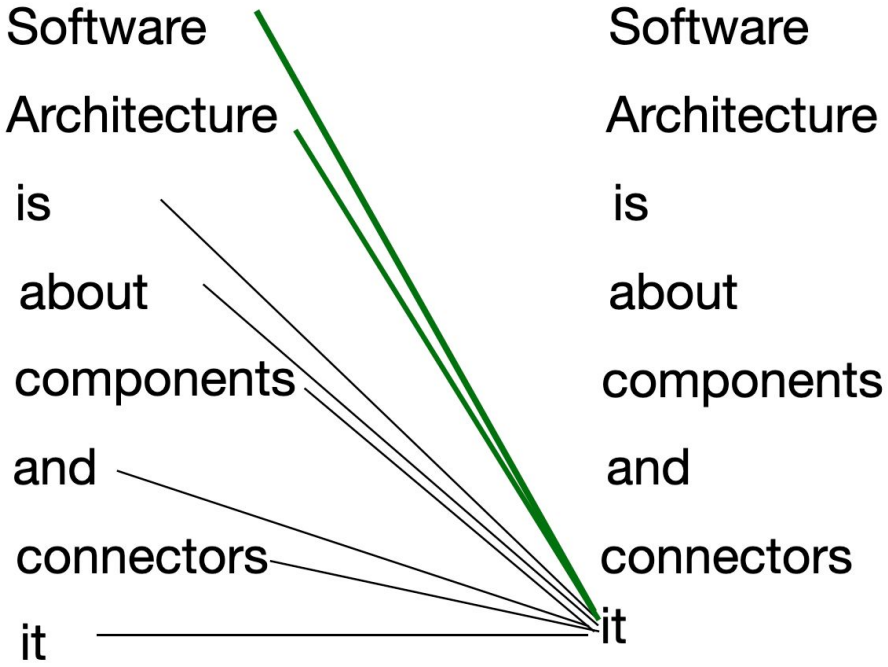
Software architecture is about components and

Software architecture is about components and connectors.

Software architecture is about components and connectors. It

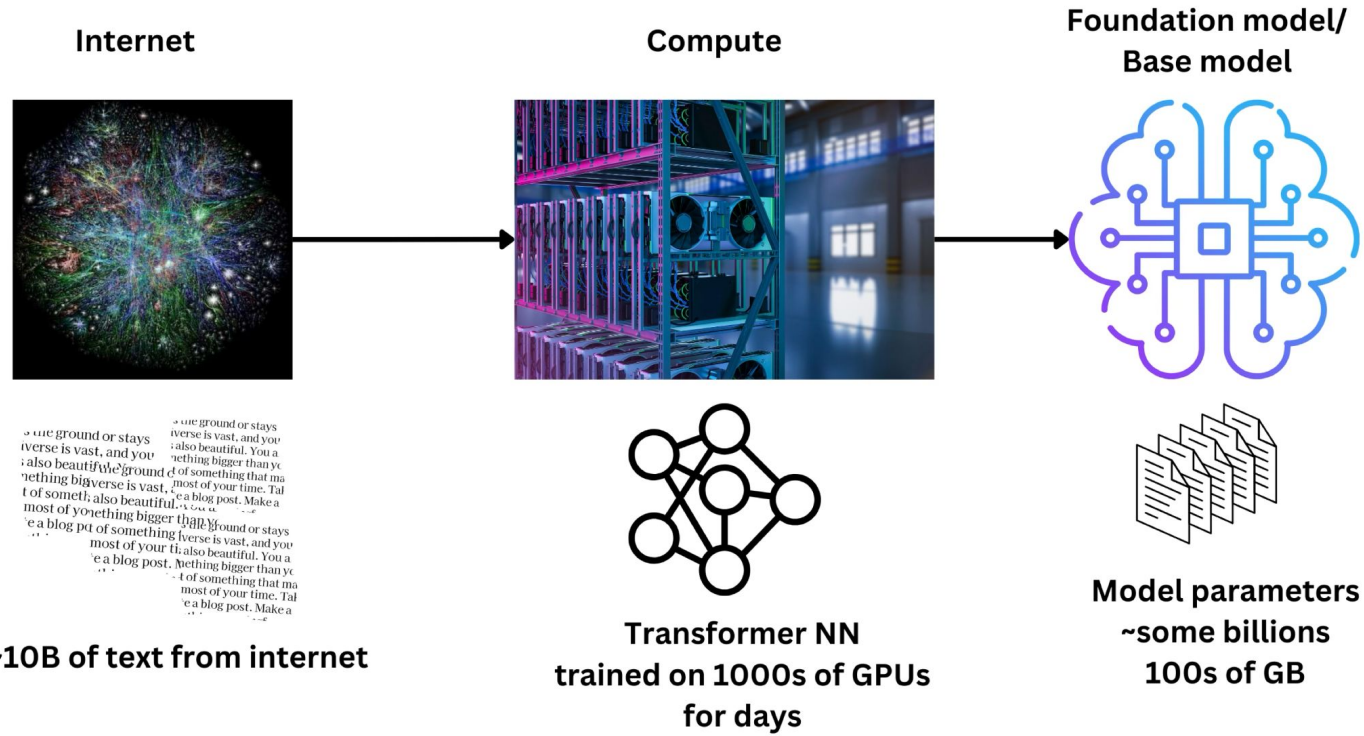
The world of Transformers





Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I., 2017. **Attention is all you need.** *Advances in neural information processing systems*, *30*.

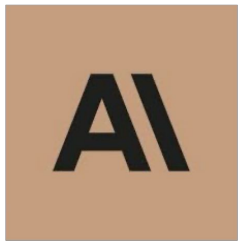
Large Language Models



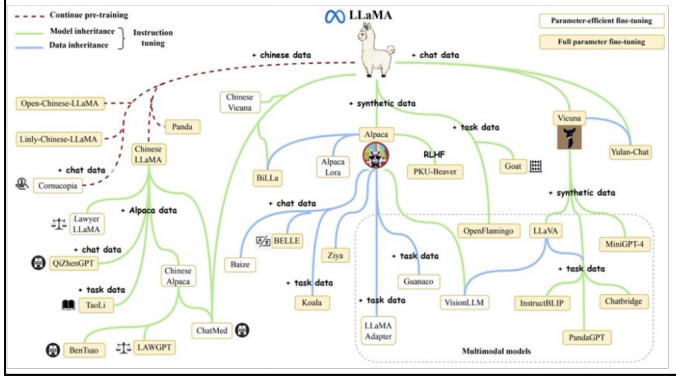
Source: Inspiration from slides of Andrej Karpathy, Introduction to large language models

Broad Landscape of LLMs

Proprietary



Open Source



Text

3 days ago

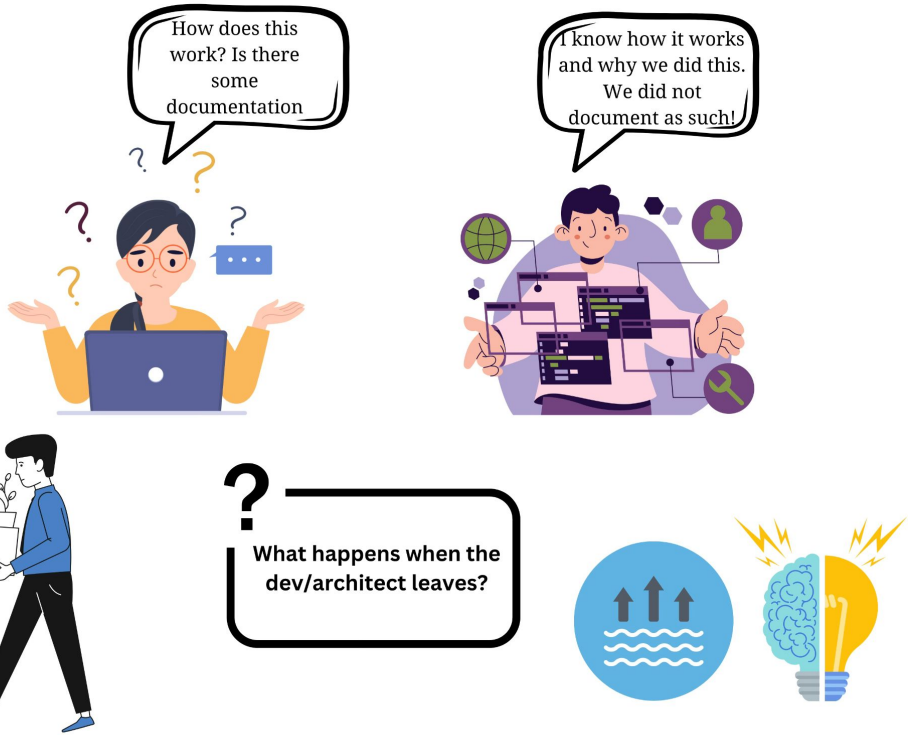
Rank (UB) ↑	Model ↑↓	Score ↑↓	Votes ↑↓
1	gemini-2.5-pro	1455	41,731
1	claude-opus-4-1-20250805-thi…	1451	11,750
2	o3-2025-04-16	1444	43,898

Copilot

105 days ago

Rank (UB) ↑	Model ↑↓	Score ↑↓	Votes ↑↓
1	Claude 3.5 Sonnet (06/20)	1012	3,544
1	Deepseek V2.5 (FIM)	1028	2,292
2	Codestral (05/24)	1001	5,744
3	GPT-4o (08/06)	986	4,464

Applying LLMs for Architecture Knowledge Management



“Software Architecture is a set of key design Decisions.”

Janson et. al
WICSA

Capturing Design Decisions Through ADRs

- **Architecture Decision Records: ADR**
- Lightweight mechanism for documenting decisions
- Design decisions require careful considerations of various parameters
  - This requires broader understanding of domain as well as expertise
  - Can LLMs generate ADRs given context due to world knowledge?

Title: Deciding the technology for the data analysis component

Context
We need to decide whether to use Python as a programming language for our project. Our project involves data analysis, machine learning, and web development.

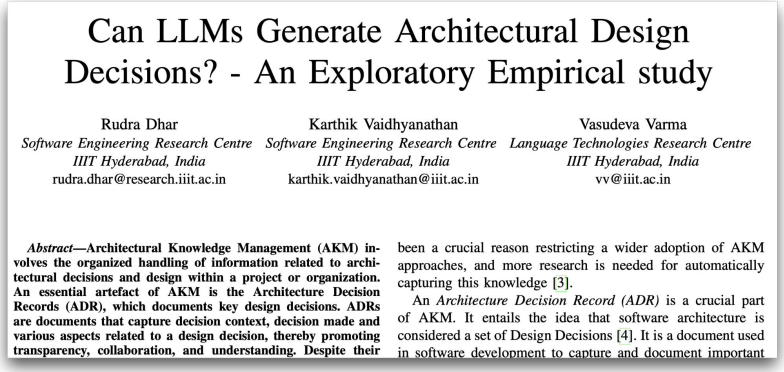
Decision
We have decided to use Python as our primary programming language for our project.

Rationale
1. Python has support for various ML and data analysis
2. Team members are already familiar with Python

Status
Decided

Consequences
1. Functionality needs to be exposed as API for integration support

Exploring the Capabilities of LLMs for Generating Design Decisions



Can LLMs Generate Architectural Design Decisions? - An Exploratory Empirical study

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Vasudeva Varma
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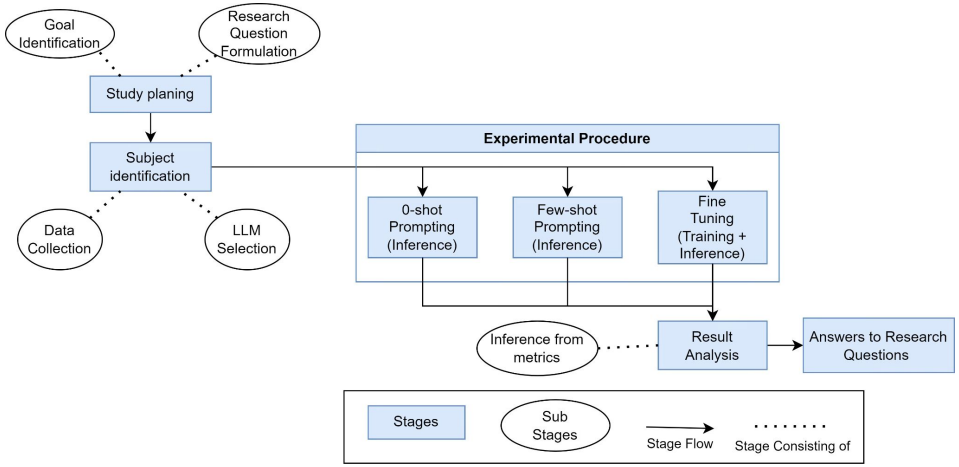
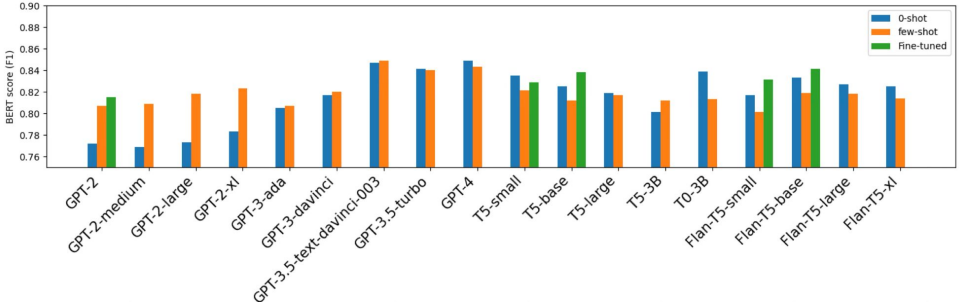
***Abstract*—Architectural Knowledge Management (AKM) involves the organized handling of information related to architectural decisions and design within a project or organization. An essential artefact of AKM is the Architecture Decision Records (ADR), which documents key design decisions. ADRs are documents that capture decision context, decision made and various aspects related to a design decision, thereby promoting transparency, collaboration, and understanding. Despite their** been a crucial reason restricting a wider adoption of AKM approaches, and more research is needed for automatically capturing this knowledge [3].

An *Architecture Decision Record (ADR)* is a crucial part of AKM. It entails the idea that software architecture is considered a set of Design Decisions [4]. It is a document used in software development to capture and document important





Fine tuned small models perform closer to large ones .

Dhar et al. ICSA 2024

Time for Hands on with LLMs

Part 1

Zero-Shot

Use simple prompting and give custom System prompts to generate ADRs for a given context

Part 2

Few-Shot

Give examples of ADRs along with System prompts

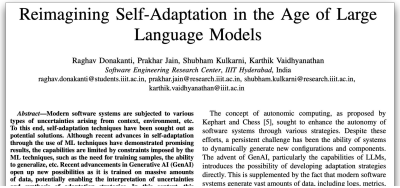
Part 3

RAG

Create a small repo of ADRs and use that to retrieve ADR for a given context

Notebook: https://tinyurl.com/ecsa2025-agentic-1

Many more studies on using LLMs for SA Tasks



Reimagining Self-Adaptation in the Age of Large Language Models

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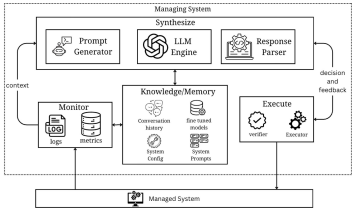
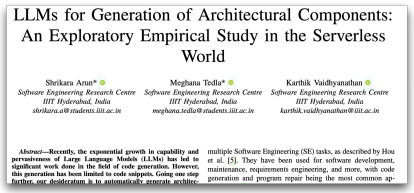
Abstract—Modern software systems are subjected to various types of uncertainties arising from context, environment, etc. To this end, self-adaptation techniques have been sought out as potential solutions. Although recent advances in self-adaptation through the use of ML techniques have demonstrated promising results, the capabilities are limited by constraints imposed by the ML techniques, such as the need for training samples, the ability to generalize, etc. Recent advancements in Generative AI (GenAI) open up new possibilities as it is trained on massive amounts of data, potentially enabling the interpretation of uncertainties

The concept of autonomic computing, as proposed by Kephart and Chess [5], sought to enhance the autonomy of software systems through various strategies. Despite these efforts, a persistent challenge has been the ability of systems to dynamically generate new configurations and components. The advent of GenAI, particularly the capabilities of LLMs, introduces the possibility of developing adaptation strategies directly. This is supplemented by the fact that modern software systems generate vast amounts of data, including logs, metrics.

ICSA-C 2024





LLMs for Generation of Architectural Components: An Exploratory Empirical Study in the Serverless World

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Abstract—Recently, the exponential growth in capability and pervasiveness of Large Language Models (LLMs) has led to significant work done in the field of code generation. However, this generation has been limited to code snippets. Going one step further, our deliberation is to automatically generate architec-

multiple Software Engineering (SE) tasks, as described by Hou et al. [5]. They have been used for software development, maintenance, requirements engineering, and more, with code generation and program repair being the most common ap-

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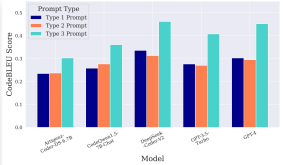

Leveraging LLMs for Dynamic IoT Systems Generation through Mixed-Initiative Interaction

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ICSA-C 2025

Towards Human-Bot Collaborative Software Architecting with ChatGPT

Aakash Ahmad[1], Muhammad Waseem[2], Peng Liang[3], Mahdi Fehmideh[4], Mst Shamima Aktar[3] and Tommi Mikkonen[2]

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[3]School of Computer Science, Wuhan University, Wuhan, China
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EASE 2023

State of Practice: LLMs in Software Engineering and Software Architecture

Publisher: IEEE Cite This PDF

Jasmin Jahić ; Ashkan Sami All Authors

ICSA-C, 2024

From Requirements to Architecture: An AI-Based Journey to Semi-Automatically Generate Software Architectures

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Workshop on Designing @ICSE 2024

ARTICLE

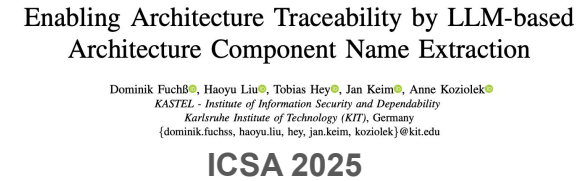
Helping Novice Architects to Make Quality Design Decisions Using an LLM-Based Assistant

Authors: J. Andrés Díaz-Pace, Antonela Tommasel, Rafael Capilla | Authors Info & Claims

Software Architecture: 18th European Conference, ECSA 2024, Luxembourg City, Luxembourg, September 3–6, 2024, Proceedings
Pages 324 - 332 • https://doi.org/10.1007/978-3-031-70797-1_21

Published: 01 September 2024 Publication History

ECSA 2024



Enabling Architecture Traceability by LLM-based Architecture Component Name Extraction

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ICSA 2025

Do Large Language Models Contain Software Architectural Knowledge?

An Exploratory Case Study with GPT

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ICSA 2025

ARTICLE

Mapping Source Code to Software Architecture by Leveraging Large Language Models

Authors: Nils Johansson, Mauro Caporuscio, Tobias Olsson | Authors Info & Claims

Software Architecture. ECSA 2024 Tracks and Workshops: Luxembourg City, Luxembourg, September 3–6, 2024, Proceedings
Pages 133 - 149 • https://doi.org/10.1007/978-3-031-71246-3_13

ECSA 2024

Around 4+ studies@ECSA 2025: Arch Knowledge, Arch smells, recovery, assessment of diagrams, cost reduction

GenAI for SA: What has been done?

Generative AI for Software Architecture. Applications, Challenges, and Future Directions

Matteo Esposito[a], Xiaozhou Li[a], Sergio Moreschini[a,b], Noman Ahmad[a], Tomas Cerny[c], Karthik Vaidhyanathan[d], Valentina Lenarduzzi[a], Davide Taibi[a]

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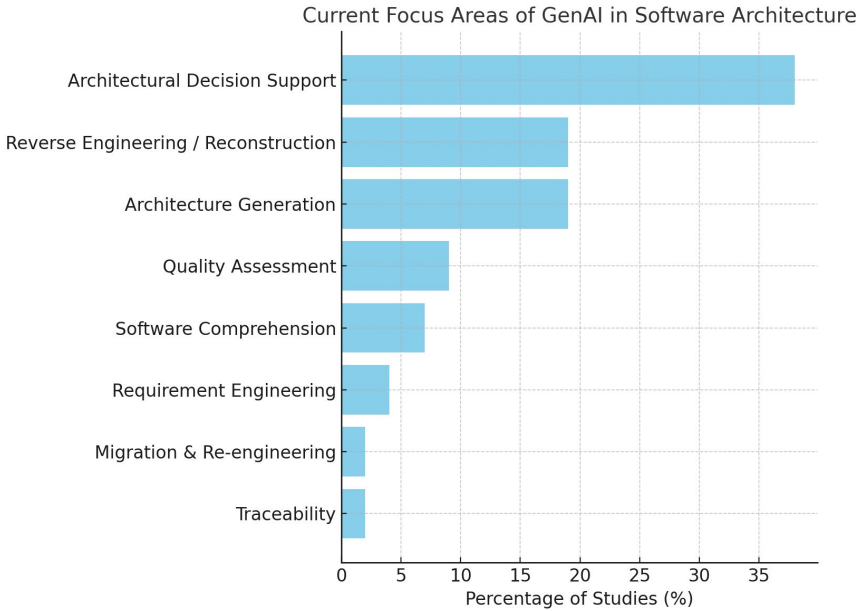
[d] *Software Engineering Research Center, IIIT Hyderabad, India*

Abstract

Context. Generative Artificial Intelligence (GenAI) is transforming much of software development, yet its application in software architecture is still in its infancy.
Aim. Systematically synthesize the use, rationale, contexts, usability, and challenges of GenAI in software architecture.
Method. Multivocal literature review (MLR), analyzing peer-reviewed and gray literature, identifying current practices, models, adoption contexts, reported challenges, and extracting themes via open coding.
Results: This review identifies a significant adoption of GenAI for architectural decision support and architectural reconstruction. OpenAI GPT models are predominantly applied, and there is consistent use of techniques such as few-shot prompting and retrieval-augmented generation (RAG). GenAI has been applied mostly to the initial stages of the Software Architecture Life Cycle (SALC), such as Requirements-to-Architecture and Architecture-to-Code. Monolithic and microservice architectures were the main targets. However, rigorous testing of GenAI outputs was typically missing from the studies. Among the most frequent challenges are model precision, hallucinations, ethical aspects, privacy issues, lack of architecture-specific datasets, and the absence of sound evaluation frameworks.
Conclusions: GenAI shows significant potential in software design, but there are several challenges on its way toward greater adoption. Research efforts should target designing general evaluation methodologies, handling ethics and precision, increasing transparency and explainability, and promoting architecture-specific datasets and benchmarks to overcome the gap between theoretical possibility and practical use.

Keywords: Generative AI, Software Architecture, Multivocal Literature Review, Large Language Model, Prompt Engineering, Model Human Interaction, XAI



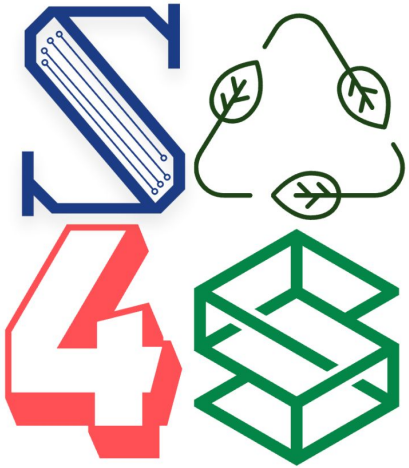
Just accepted at JSS

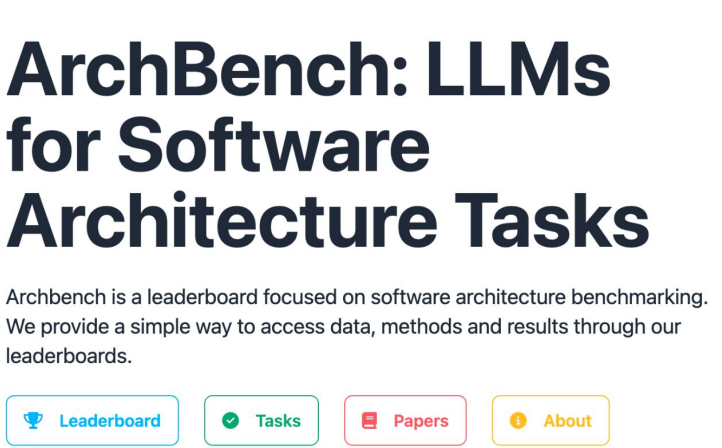
GenAI for SA: Lot more needs to be done!

- **Lack of benchmarks and datasets** in Software Architecture Community
- Need for more **robust evaluation** strategies - Implications on real-world
- Move towards more complex architectures and consider heterogeneous inputs
- We may have to rethink the architecting process - **Human + AI cooperation**
- **Ethical considerations, privacy, IP** - Need more answers here!
- Shift towards more **domain specific LLMs (SLMs)** can be also way forward!

Software Architecture Bench - sabench

- Collection of three tasks
  - ADR
  - Serverless function generation
  - Dynamic service generation
- Datasets and leaderboards
- You can contribute as well!
- **Feedbacks welcome!**





From Text Generation to Action: Into the World of Agentic AI

What is the latest news as of today?

Based on the provided context, I cannot give you "the latest news as of today" because this document is a report containing data and outlooks, not a real-time news source. The most recent data incorporated into the report is **up to December 20, 2023**, with some sections providing an **outlook for 2024**.

Gemini

Create a PlantUML component image for my banking system login service.....

LLMs are models at the end of the day. They can generate text but they cannot perform a function call (invoke an external tool) by themselves. We need to engineer around the model

Part 4: Agentic SA

LLM Agents - Why does it matter?

LLM-based Agents are extremely important for practitioners

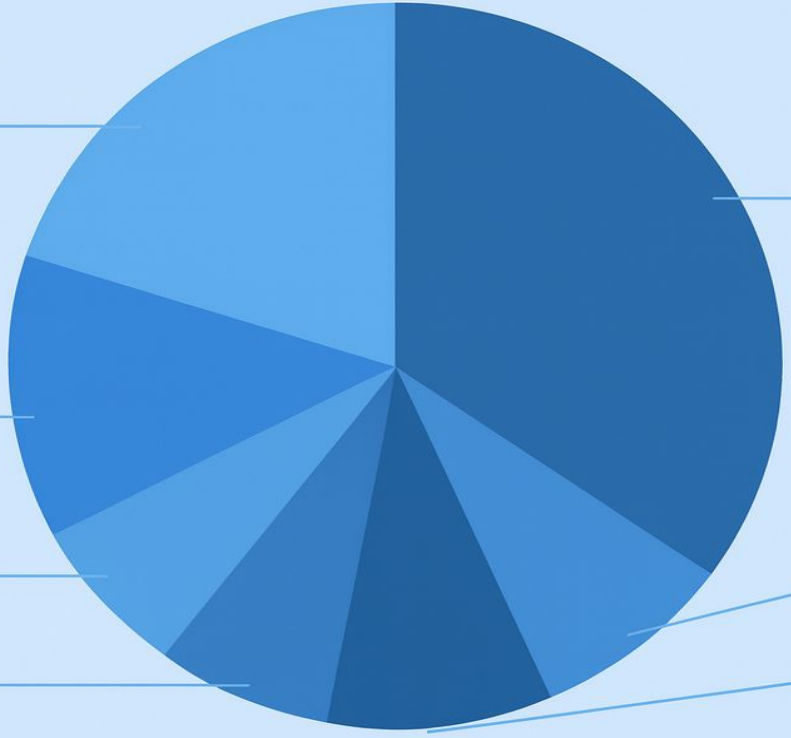
LLM vs LLM-agents in Software Engineering

LLM	LLM-agents
AI research	Software Engineering research
LLM for Software Engineering	Software Engineering for LLM

vs

[from ICSE 2025 Opening]

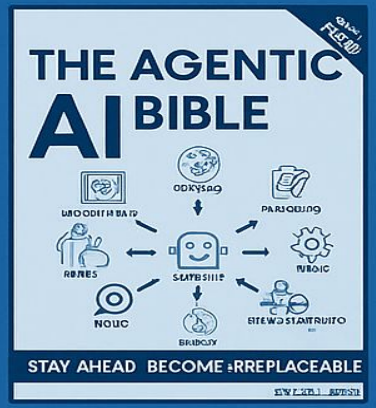
Submissions by Primary Area

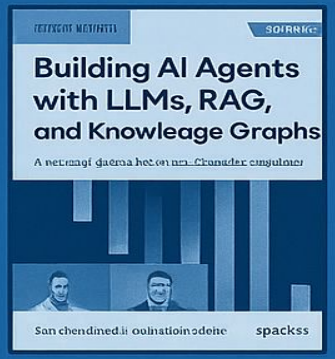


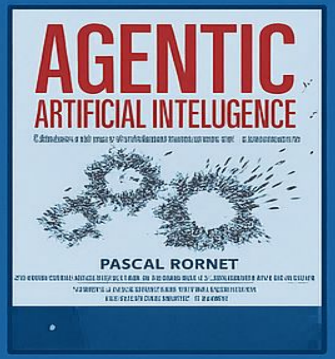
AI Agent Launches

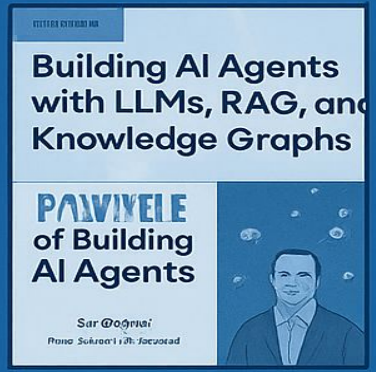
ChatGPT	July 17
Gemini	August 15
Claude	April 10
Cursor	April 18

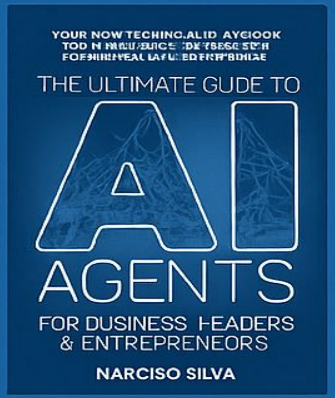
Academia is staying behind industry: again!











Disclaimer

Disclaimer

- Hype becomes harmful when we start *"... making important strategic decisions based on an incomplete and potentially flawed understanding of the core capabilities of the hyped innovation and its broader ramifications"*

- https://ieeexplore.ieee.org/document/10884672

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Editor: John Favaro
Trust-IT
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The Effects of Hype in the Software Domain: Causes, Consequences, and Mitigations

Manfred Broy and Bran Selić

From the Editor

Does the arrival of an exciting new technology too often draw our attention and resources away from the foundational pillars of our software engineering discipline that we have so painstakingly built up over decades? This month, our authors examine this question with a set of clear, well-grounded observations.—*John Favaro*

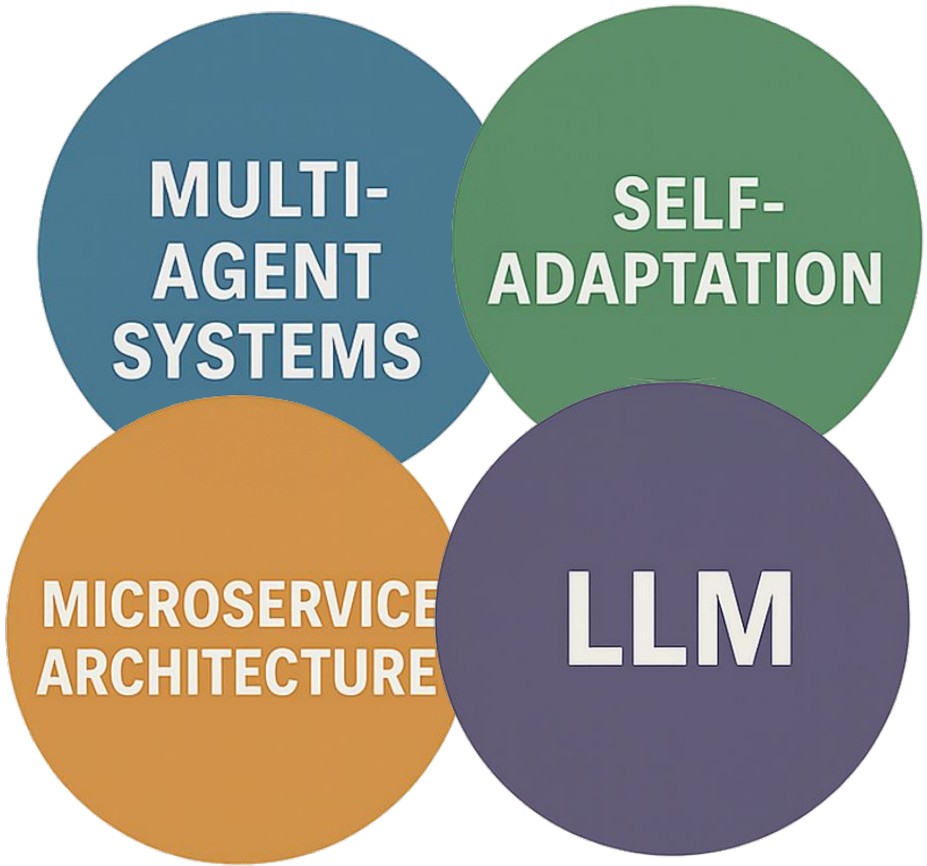
The four pillars of LLM-based Agents

MAS: 80s

Self-Adaptation in SE: late 90s

Microservice Architectures: 2010

LLM in SwEng: 2020



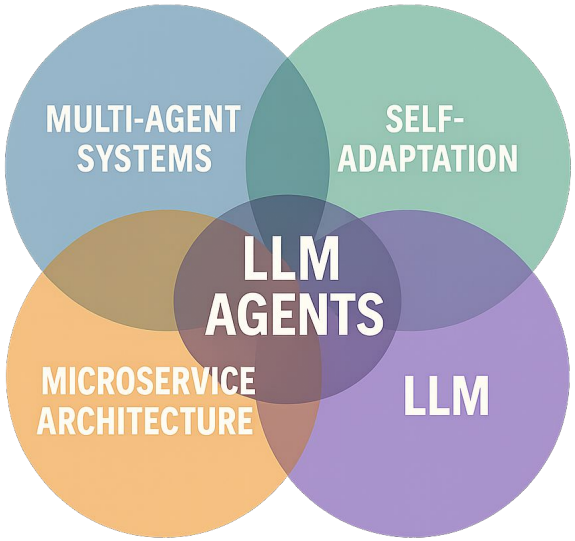
Summarizing

MAS: autonomous **entities**, working **collaboratively**, and making **autonomous** decisions.

Self-Adaptation: the ability to **autonomously** adapt based on **contextual** information.

Microservice-based Architecture: decomposition of complex systems into **“entities”** and **MS patterns**.

LLM: ability to “create” **knowledge** from existing knowledge.



What is an LLM agent?

LLM Agents

*"LLM agents are **intelligent** entities powered by large language models (**LLMs**) capable of **perceiving** environments, **reasoning** about goals, and **executing** actions"*

- Xi, Z., Chen, W., Guo, X. *et al.* **The rise and potential of large language model based agents: a survey**. *Sci. China Inf. Sci.* **68**, 121101 (2025). https://doi.org/10.1007/s11432-024-4222-0

*"A Generative AI agent can be defined as an application that attempts to achieve a **goal** by **observing** **the world** and **acting** upon it using the **tools** that it has at its disposal. Agents are **autonomous** and can act **independently of human** intervention, especially when provided with proper goals or objectives they are meant to achieve. Agents can also be **proactive** in their approach to reaching their goals"*

- Julia Wiesinger, Patrick Marlow and Vladimir Vuskovic. **Agents**. Google whitepaper (https://ppc.land/content/files/2025/01/Newwhitepaper_Agents2.pdf)

LLM Agents

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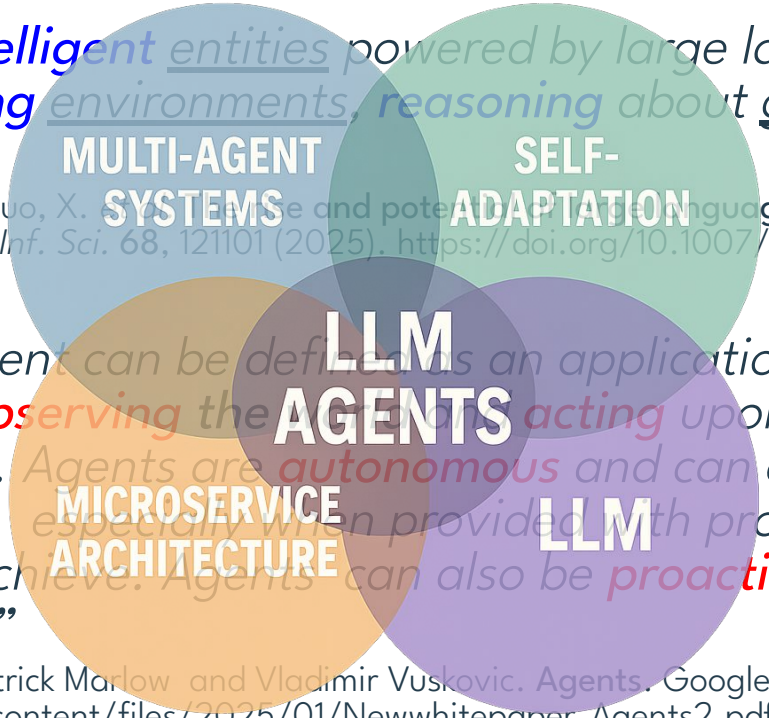


How LLM agents differ from LLMs?

An LLM is a model, an agent is an ecosystem!!

- In the context of AI, an ecosystem refers to a network of interconnected AI agents, each with specific roles and capabilities, interacting with each other and the environment.
  - Agents can collaborate, share information, and leverage each other's strengths.

LLM vs LLM Agents

LLM (Large Language Model)

- **Stateless:** No persistent memory beyond context window
- **Reactive:** Responds only to the input prompt
- Single-turn **reasoning:** Produces text in one shot
- **General-purpose** knowledge from **broad** corpora
- Limited **actionability:** outputs text only
- Opaque decision process, hard to explain reasoning

LLM Agent

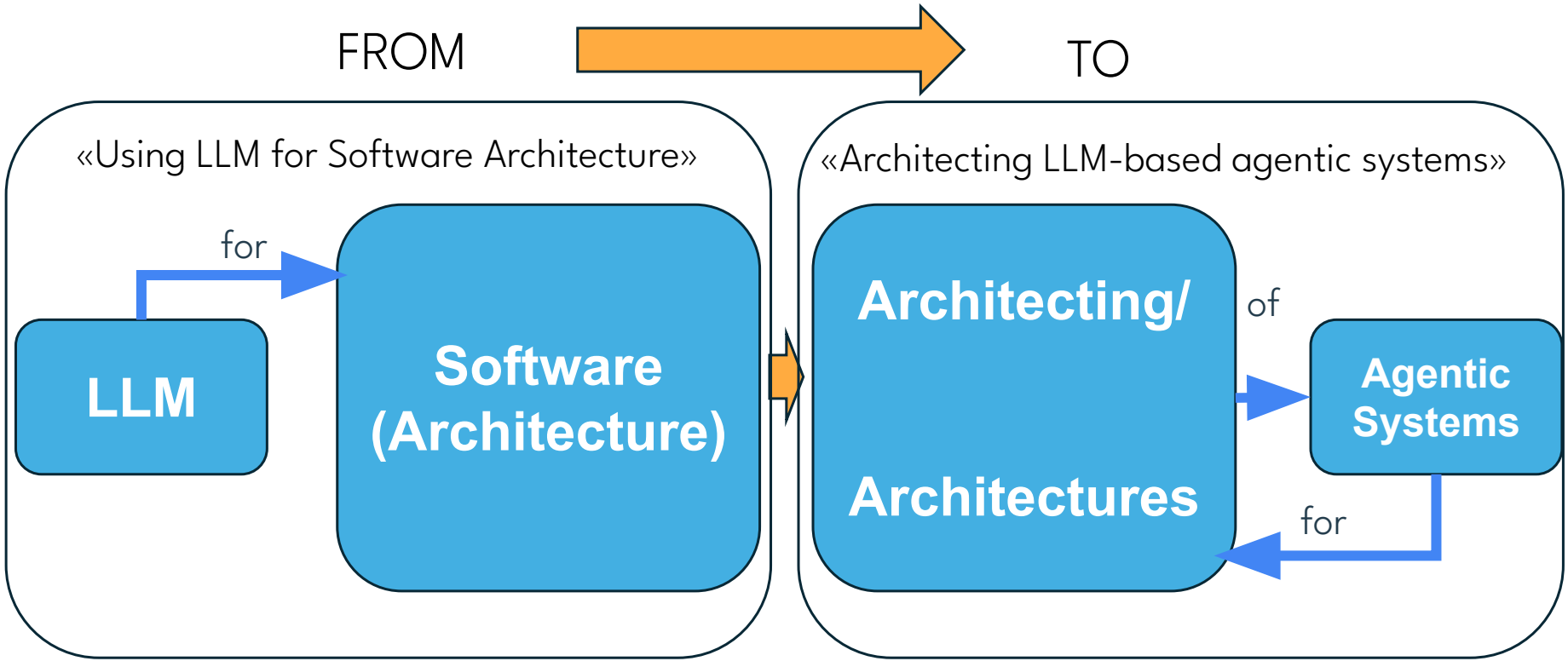
- **Stateful** with **memory:** maintains short/long-term memory
- **Proactive** & **goal-driven:** decomposes and pursues objectives
- Multi-step **reasoning** loop with **planning** & **reflection**
- **Tool-using:** invokes APIs, DBs, or external systems
- Embodied in **workflows:** coordinates tasks and collaboration
- Traceable & observable: actions logged and auditable

LLM Agents main features

LLM Agents

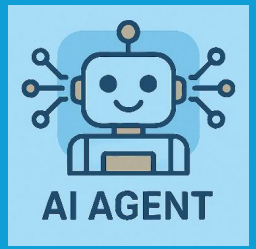
Architecting LLM agents

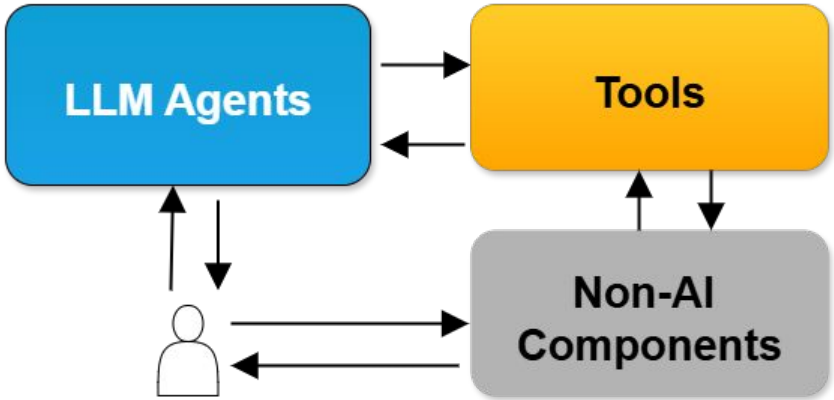
Good news: from LLM4SA to SA4LLMAgents



A single Agent Architecture – High level

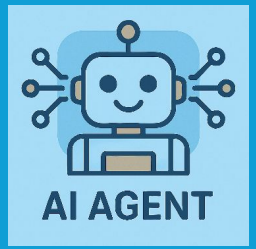
- User
- LLM Agent
- Non-AI components
- External tools

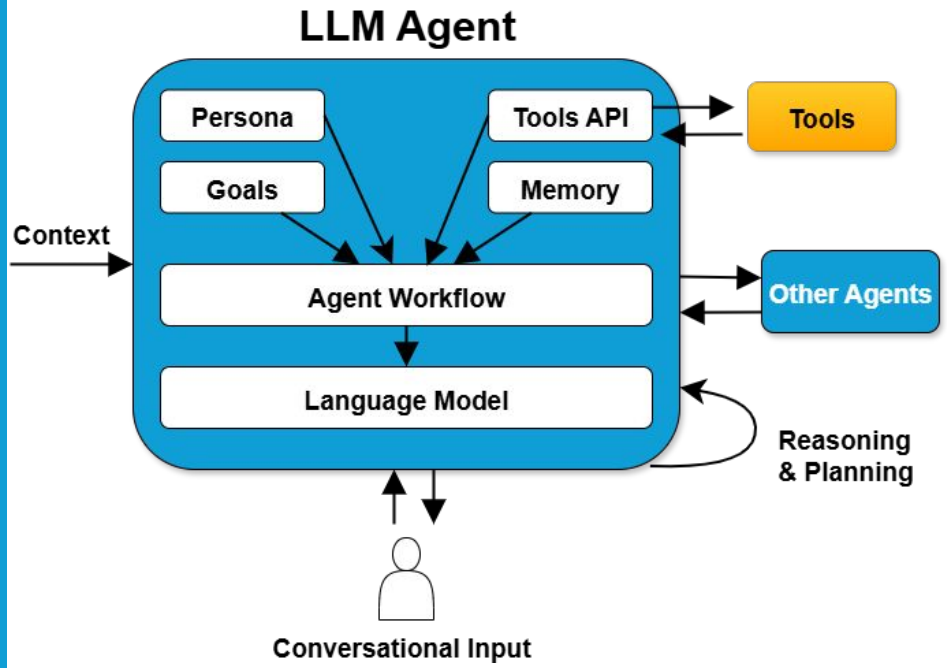




A single Agent Architecture

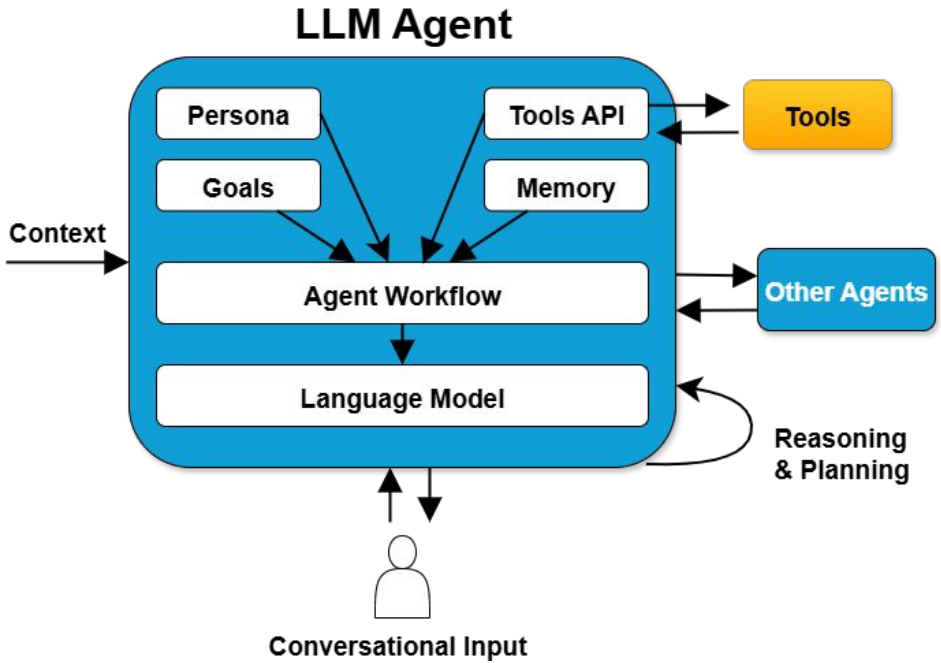
- Language Model = Brain
- Agent Workflow = Coordinator
- Memory = Short/Long-term
- Tools = Action Interfaces
- Goals = agent goals
- Persona = working context





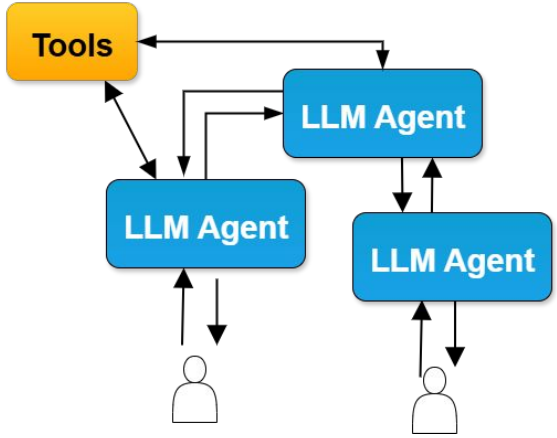
LLM agent: foundational architecture design decisions

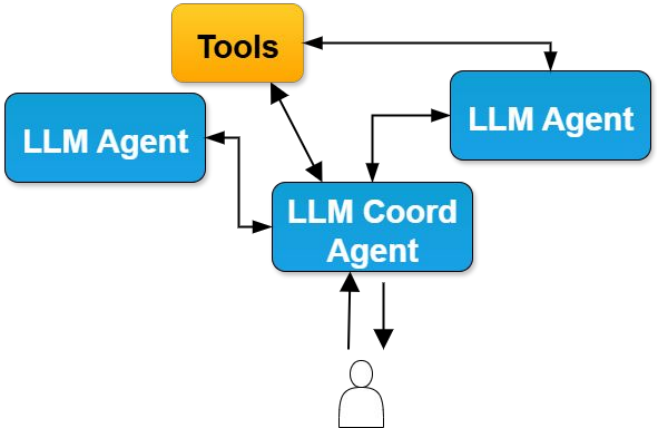
- Memory mechanisms
- Tools
- Planning and reasoning capability
- Large vs Domain-specific Language Models
- Self-Reflection, Adaptation, and Explainability



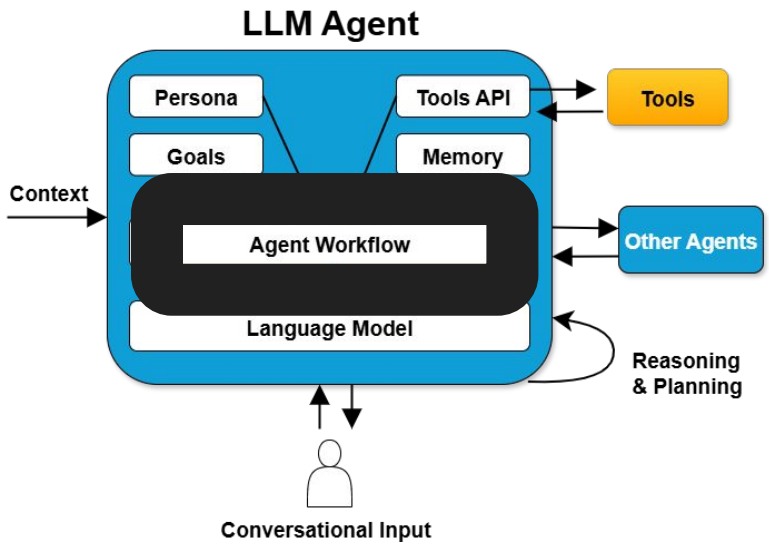
A multi-Agent LLM Architecture

- **Choreography vs Orchestration** (aka, Horizontal vs Vertical)
- **Role of Agent Workflow/coordinator**
- intelligent **division of labour** based on skill and helpful feedback from a variety of agent personas





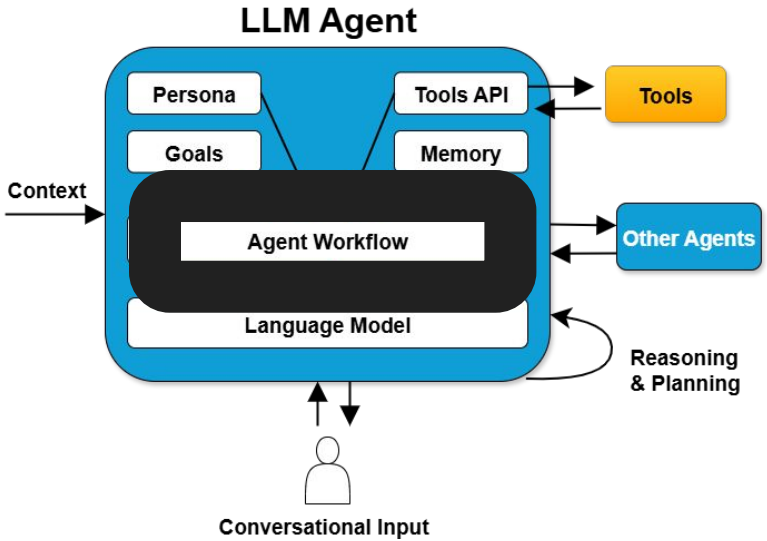
The Agent Workflow Component



It is the **"architectural spine"** of an LLM agent.

- It is one of the most architecturally significant components
- It governs ***what happens when*** in the reasoning–acting cycle
- It is **where ADD** (about planning, tool-use strategies, memory policies, and collaboration) **happens**.

The Agent Workflow Component



Core responsibilities

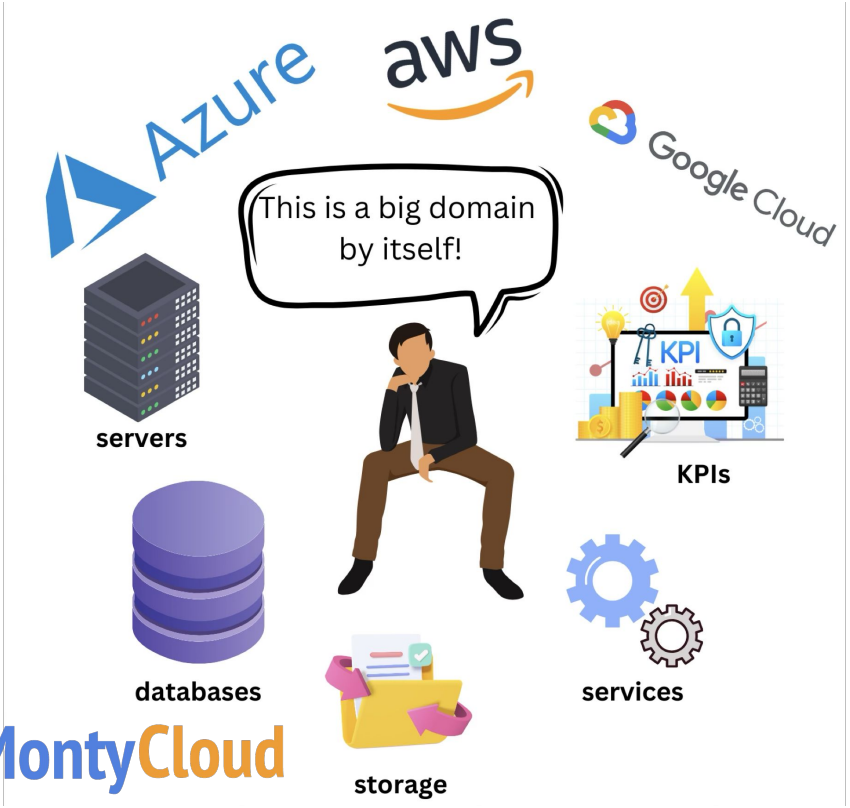
- Task Decomposition & Coordination
- Reasoning & Planning Loop
- Tool Invocation Management
- Memory & Context Handling
- Self-Monitoring
- Multi-Agent Collaboration& Reflection

Notes:

Note 1: Most of the papers on LLM agents @ICSE2025 have been about «using» multiple AI agents, rather than discussing how to create appropriate multi-agent architectures

Note 2: Most of the papers on LLM agents @ICSE2025 uses some form of «Coreography» rather than a dedicated «Agent Workflow Component" orchestrator

Agentic AI System: The case of Autonomous CloudOps



CloudOps - *Run, Manage, Evolve*

AWS Well Architected Framework

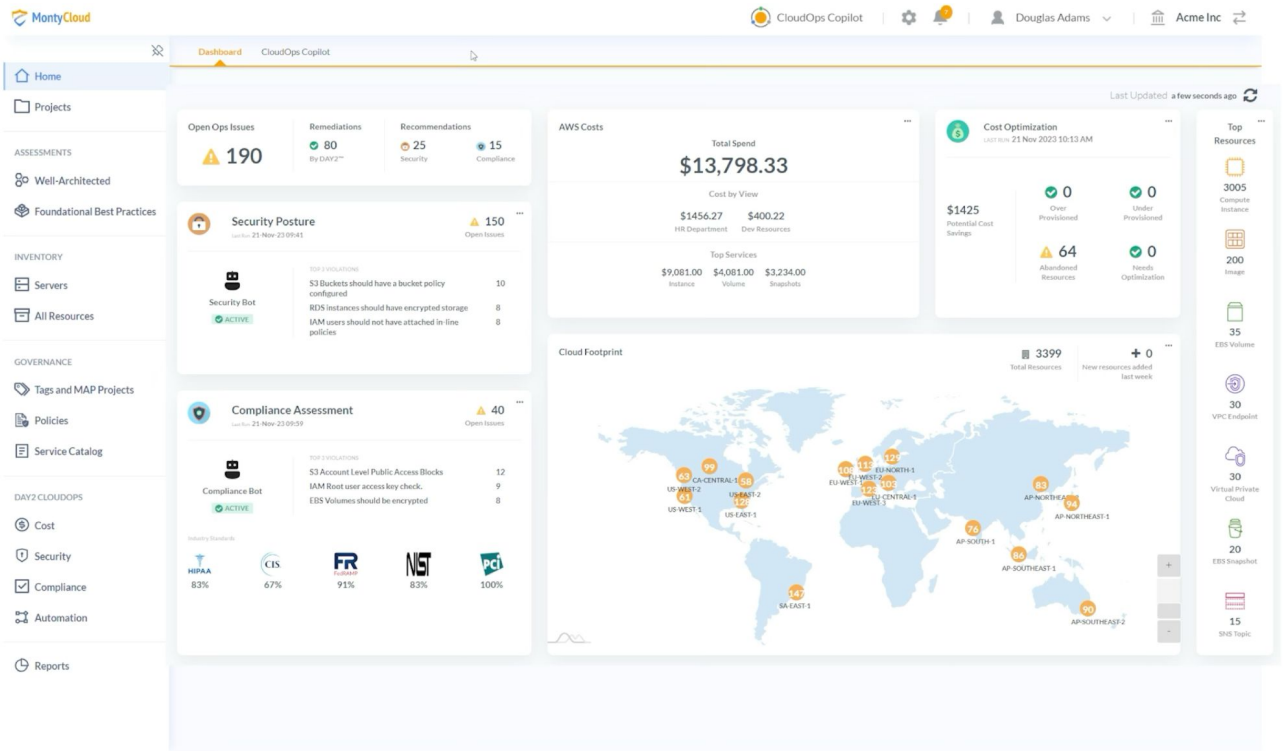
Helps cloud architects build resilient, secure and high performing infra

Build around Six Pillars

- Operational Efficiency
- Security
- Reliability
- Performance Efficiency
- Sustainability
- Cost

aws

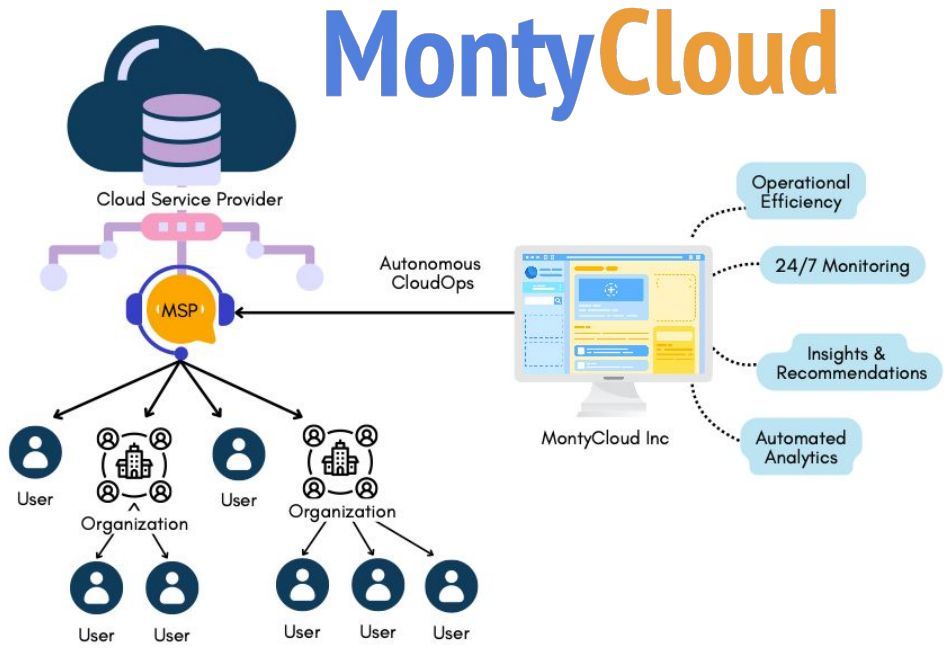
Initial development of CloudOps CoPilot



Work done in collaboration with MontyCloud Inc.

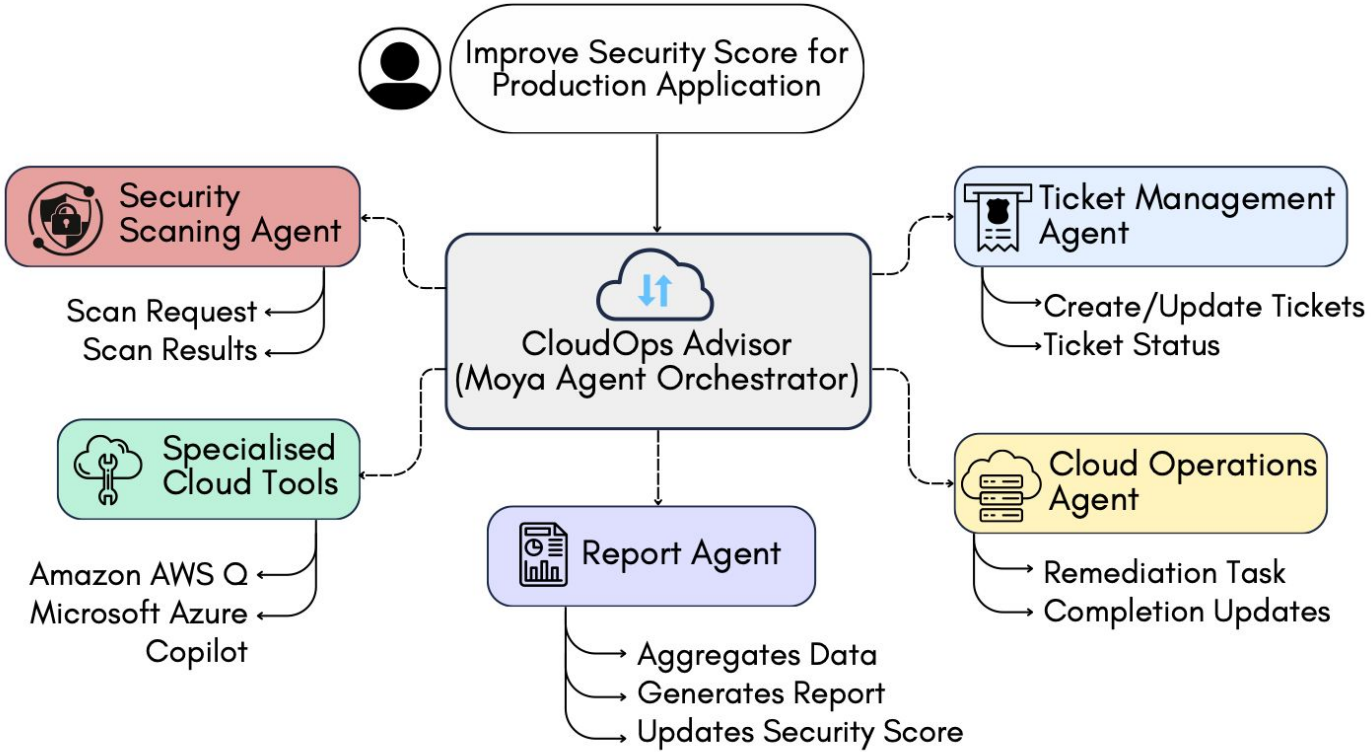
Complex Engineering Challenges

- **Managing Distributed Data**
  - Diverse data sources
- **Maintainability**
  - Large code base, time for updates
- **Extensibility and Modularity**
  - Single vendor, ease of extensions!
- **Monolithic nature of existing frameworks**
  - Limited support, vendor lock-in, learning curve



https://montycloud.com/

Going multi-agent

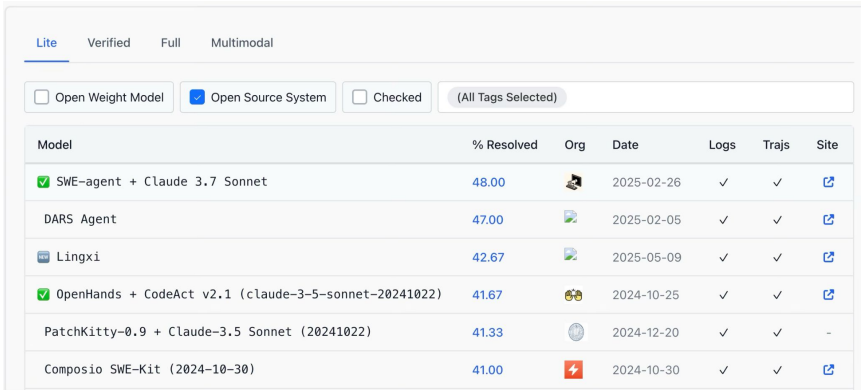


Landscape of Agentic AI Frameworks (At that point)

LangGraph

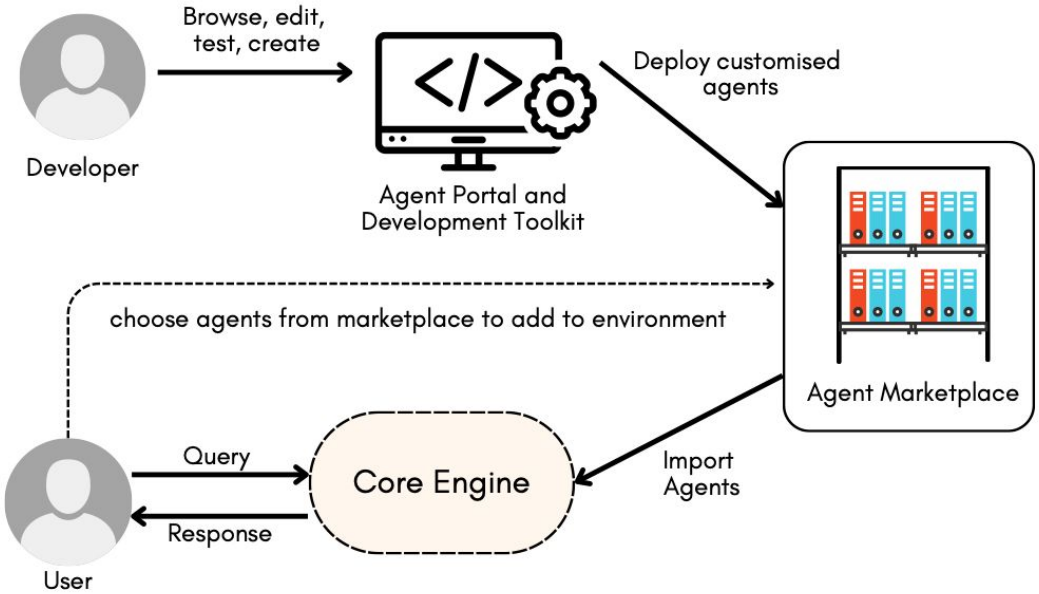
AG

crewai



Frameworks lacked flexibility and support

MOYA: Meta Orchestrator for Your Agents



Engineering LLM Powered Multi-agent Framework for Autonomous CloudOps

Kannan Parthasarathy*, Karthik Vaidhyanathan†, Rudra Dhar†, Venkat Krishnamachari*, Basil Muhammed*, Adyansh Kakran†, Sreemaee Akshathala†, Shrikara Arun†, Sumant Dubey*, Mohan Veerubhotla*, Amey Karan†
*MontyCloud Inc
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{rudra.dhar, adyansh.kakran, sreemaee.akshathala, amey.karan}@research.iiit.ac.in, shrikara.a@students.iiit.ac.in

***Abstract*—Cloud Operations (CloudOps) is a rapidly growing field focused on the automated management and optimization of cloud infrastructure which is essential for organizations navigating increasingly complex cloud environments. MontyCloud Inc is one of the major companies in the CloudOps domain that leverages autonomous bots to manage cloud compliance, security, and continuous operations. To make their platform more accessible and effective to the customers, MontyCloud worked with us to leverage the use of GenAI.**

Developing a GenAI-based solution for autonomous CloudOps for the existing MontyCloud system presented us with various challenges such as i) diverse data sources; ii) orchestration of multiple processes and iii) handling complex workflows to automate routine tasks. To this end, we developed MOYA, a multi-agent framework that leverages GenAI and balances autonomy with the necessary human control. This framework integrates various internal and external systems and is optimised for factors like task orchestration, security, and error mitigation while producing accurate, reliable and relevant insights by utilising Retrieval Augmented Generation (RAG). Evaluations of our multi-agent system with the help of practitioners as well as using automated checks demonstrate enhanced accuracy, responsiveness, and effectiveness over non-agentic approaches across complex workflows.

a well-defined CloudOps practice to effectively manage their share of duties.

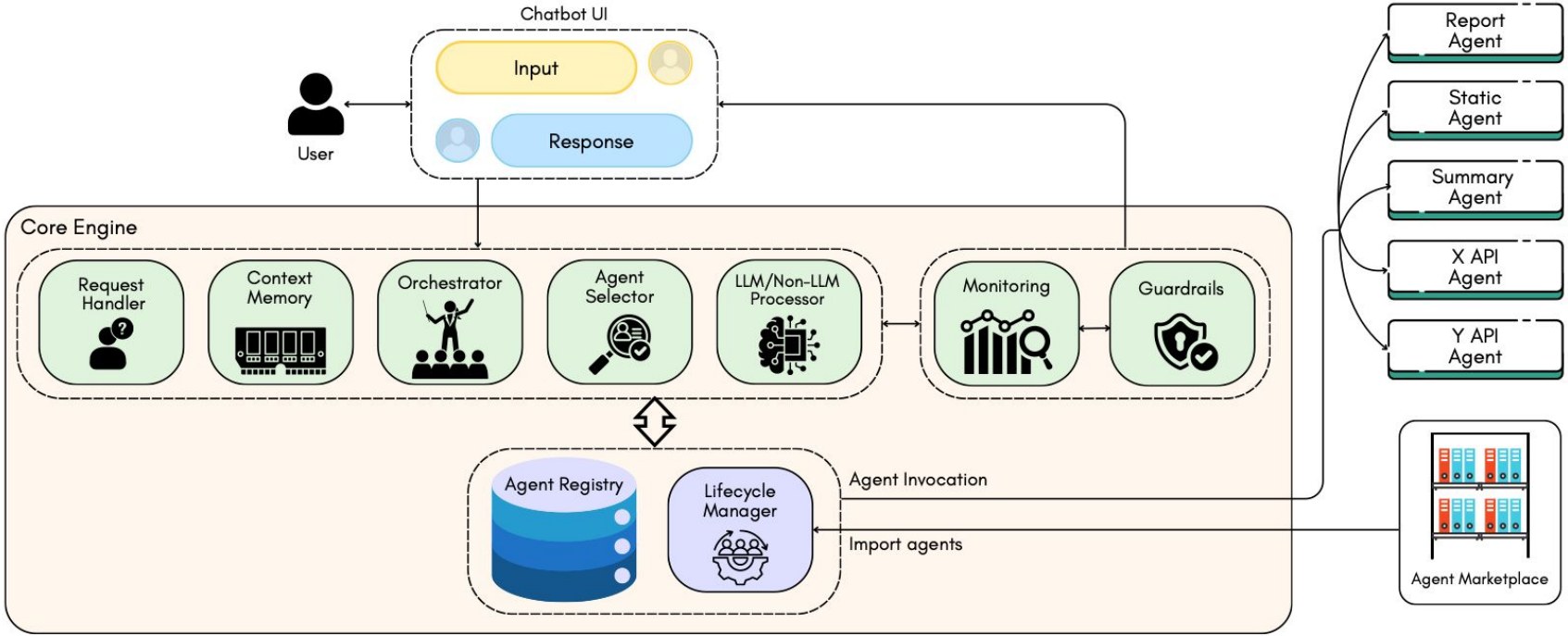
CloudOps, or Cloud Operations, refers to the practices, tools, and processes to manage, optimise, and secure applications and infrastructure in the cloud. Alonso et al. [1] defines it as a framework that extends *DevOps* practices to cloud management by adding components like resource discovery, self-healing, and real-time monitoring. By focusing on automation, monitoring, cost management, and compliance, CloudOps enables organisations to maintain efficient, resilient, and scalable cloud environments. However, the complex and dynamic nature of cloud services makes manual management time-intensive, challenging, and prone to errors.

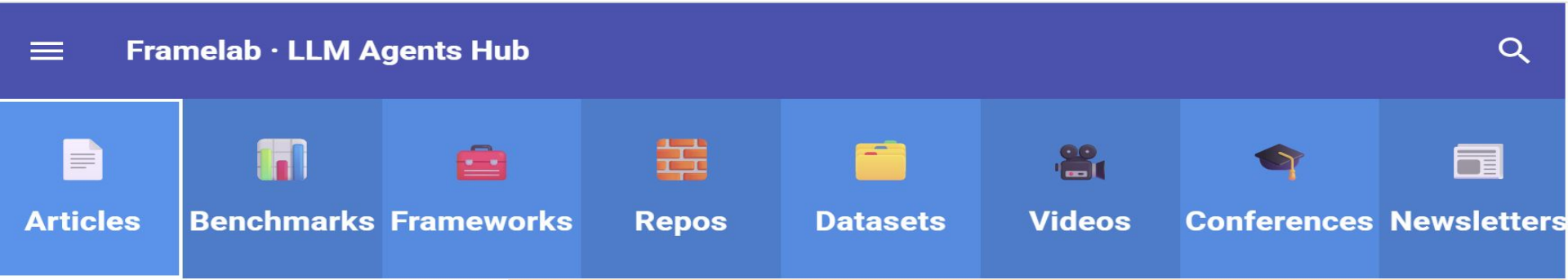
MontyCloud's Autonomous CloudOps platform addresses these challenges by automating workflows to streamline operations and provide real-time visibility into inventory, security, and costs [4]. The platform tackles challenges such as navigating the complexity of hundreds of services, establishing secure and cost-effective cloud governance, ensuring a strong security posture, and adhering to evolving compliance standards.

Best paper candidate@CAIN 2024!

Architecture of MOYA

Moya Repo





https://framelab-univaq.github.io/llm-agents-hub/

Research Papers

Peer-reviewed papers and surveys on agentic LLMs. The table lists Title, Year/Venue, a one-line Takeaway, and a Link (PDF/project).

Title	Year	Venue	Notes
Learning to Use Tools via Cooperative and Interactive Agents	2024	arXiv	Tool use, multi-agent
Reflexion: Language Agents with Verbal Reinforcement Learning	2023	arXiv	Self-reflection, RL
Systematic Biases in LLM Simulations of Debates	2024	arXiv	Simulation bias
Evil Geniuses: Delving into the Safety of LLM-based Agents	2024	arXiv	Safety

Time for Hands on with LLM Agents

Part 1

Tool Augmented Agents

LLMs uses tools to perform a given task

Part 2

Orchestrated

Multiple agents collaborate together to book a trip to Cyprus. The coordination is performed by an orchestrator component

Part 3

Multi-agent

Multiple agents dynamically collaborate without a central orchestrator using a blackboard like pattern

Notebook: https://tinyurl.com/ecsa2025-agentic-2

Architecting LLM Agents: State of the Art and What is Missing (Overview)

Software Architecture @SWEBOK v4.0

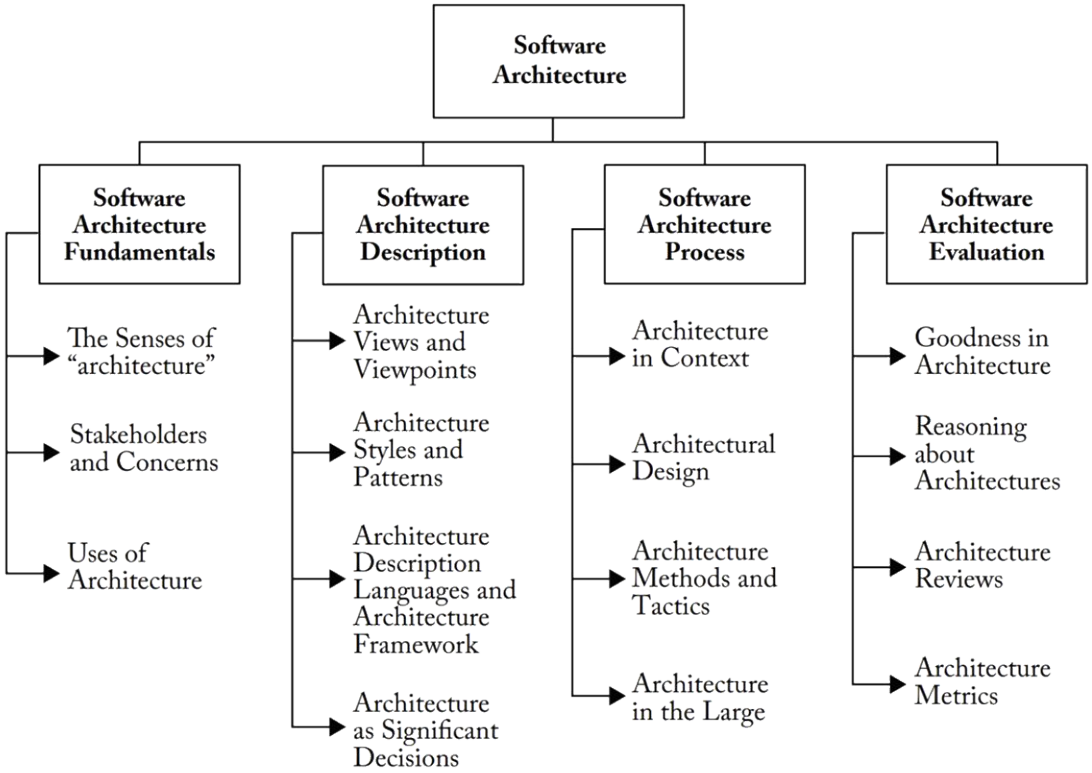


Figure 2.1. Breakdown of Topics for the Software Architecture KA

SoTA on Agentic System Architecture @SWEBOK v4.0

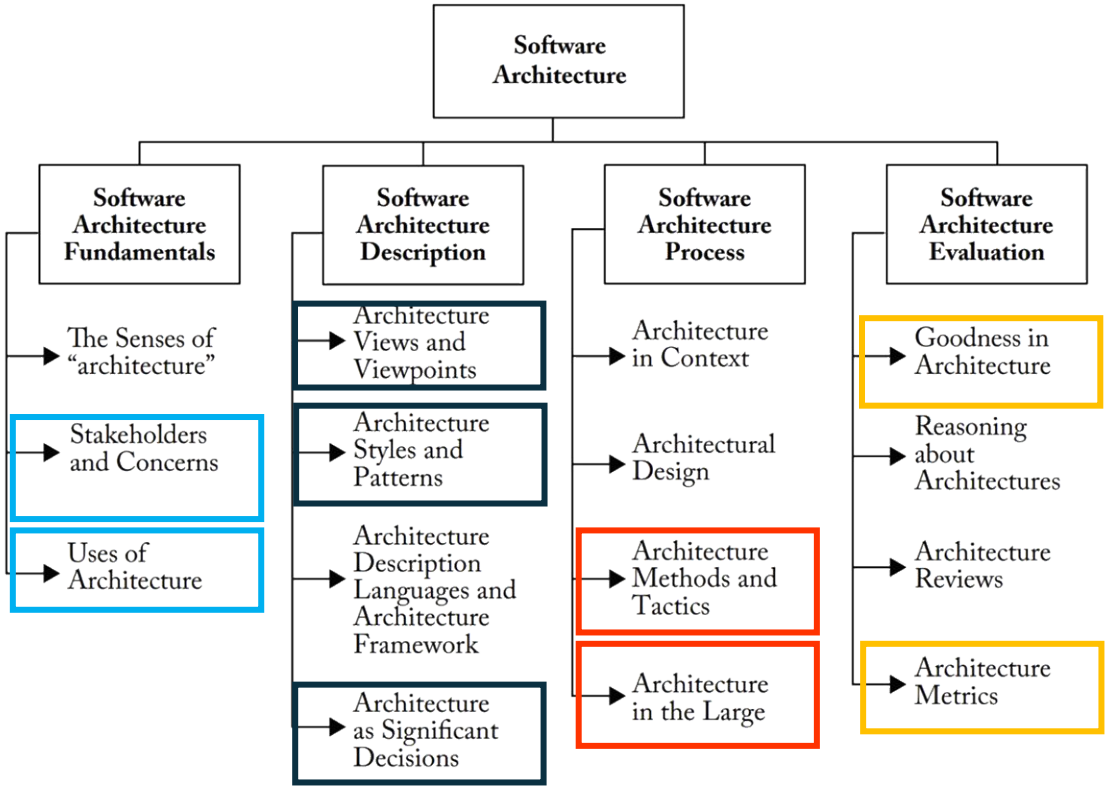
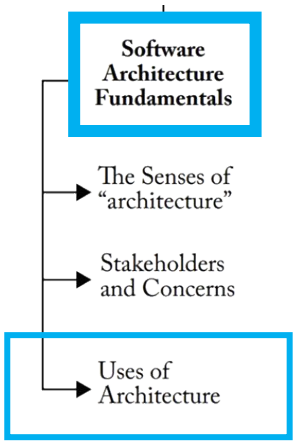


Figure 2.1. Breakdown of Topics for the Software Architecture KA

What the study is about



- Overall Statement about the SoTA

  Architectural descriptions in this domain are used as **templates or guides** for building new agents.

- Literature review on the topic (with references)

  - *Reference architecture* for FM-based agents [Lu et al. 2024]
  - A *taxonomy* (construction, collaboration, evolution) to **organize LLM-agent systems** [Luo et al. 2025]

- "What is missing"

What is Missing

- Ok as a blueprint, but **little** on using architecture for **project communication** or as a **contract** between teams.
- No formal **architecture documentation** (possibly because of small enough agentic systems so far)

Main findings

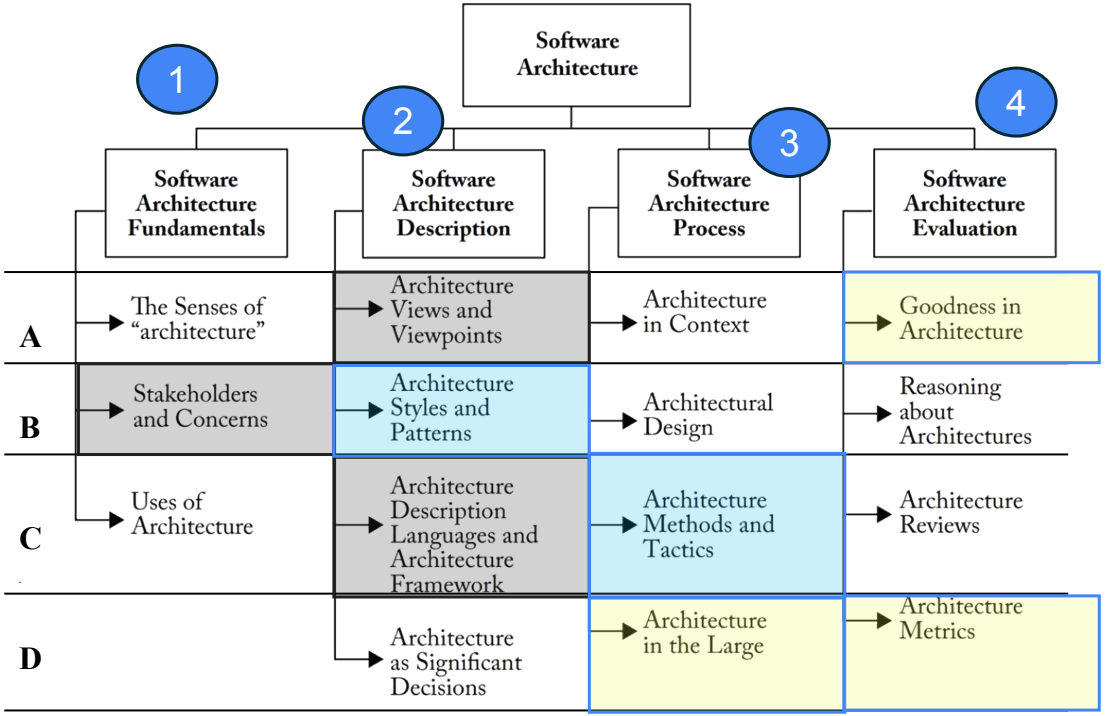


Figure 2.1. Breakdown of Topics for the Software Architecture KA

Main finding: Quality of large Agentic systems

Ref to SWEBOK	**3D, 4A, 4D**
Topic	Evaluating Large Architectures
SoTA	The question of scaling agent architectures to larger systems or organizations is **just beginning to be raised.** A number of **qualities** have been mentioned in articles.
What is Missing	Metrics to **evaluate** complex agent-based architectures are still missing. They are particularly relevant for **large** architectures that may contain hundreds of different LLM agents. We also lack quantitative **non-functional requirements evaluation, as well as trade-off analysis,** for agent architectures

Main finding: Style, Patterns, and Tactics

Ref to SWEBOK	**1B, 2A, 2C**
Topic	Architecture Style, Patterns, and Tactics
SoTA	**Numerous** ***design patterns*** and ***styles*** for LLM-based agents have been identified (e.g., Single vs. Multi-Agent, Reasoning/Planning Patterns,Memory and Tool-Use Integration) A **rich variety** of ***tactics*** and methods have been proposed as well.
What is Missing	While many (architectural) patterns have been proposed, as well as tactics, they are presented ***in isolation***, missing an overall «system» view, and mostly **focusing on the LLM-agents sub-systems**, rather than the full scope.

Main finding: Architecture Description

Ref to SWEBOK	**2B, 3C**
Topic	Architecture Description, Stakeholder, Concerns, View and Viewpoints
SoTA	Work on LLM agents **implicitly** describes **certain** views. A few **stakeholders** and **concerns** are also identified.
What is Missing	An **Architecture Framework**, according to the IEEE/ISO/IEC 42010 standard, is still **missing**. Stakeholders and Concerns are only briefly and **informally** discussed. How stakeholder concerns are reflected in architecture descriptions is missing. An architecture description language for analysis purposes is still missing.

Beyond the Horizon

Food for Thoughts

Research opportunities [ROs]

- [RO1] Skills and Competencies
- [RO2] Engineering Large-scale MAS
- [RO3] Energy Consumption
- [RO4] Agentic Systems Quality
- [RO5] Agentic Systems Engineering in Practice
- [RO6] Architecting Agentic Memory
- [RO7] Agentic Self-adaptation
- [RO8] Legacy to Agentic Modernization

Food for Thoughts > [RO1] Skills and Competencies

Image from https://oxford-management.com/

What skills will software engineers need in the Agentic AI era?

- How shall academia **train** a new generation of software engineers?
- What are the main **challenges** a Software Engineer should know when being trained on Agentic AI Engineering?
- Which are the competencies required by **the industry when** transiting to Agentic Systems Architecting?
- etc.

Food for Thoughts > [RO2] Engineering Large-scale MAS

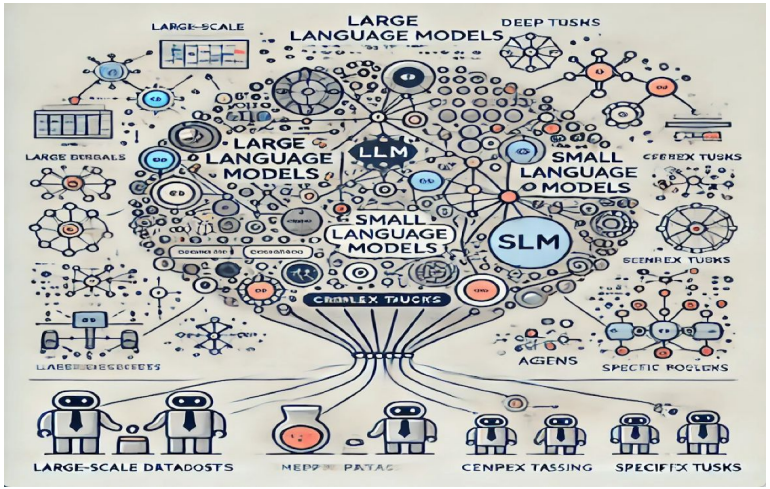
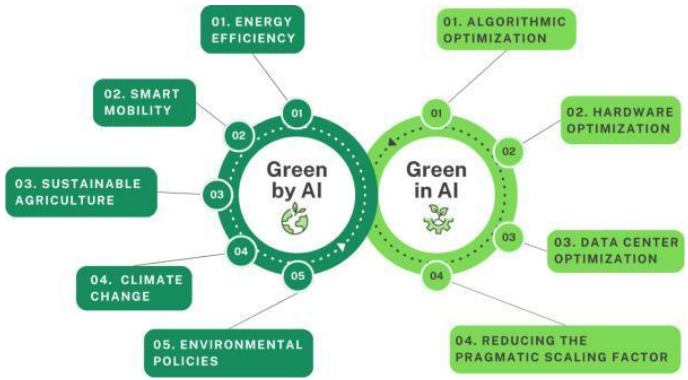


Image from https://siliconangle.com/

How to Engineer an Agentic System with hundreds of collaborating agents?

- How to **distribute roles** to agents?
- How to **"monitor"** the behavior of collaborating agents?
- How to manage **repositories** of agents?
- Agents' **selection** mechanisms
- etc.

Food for Thoughts > [RO3] Energy Consumption



How to deal with Green AI in the era of Agentic Systems?

- **Large** Language Models vs **Small** Language Models
- **Energy-Aware** design of Agentic Systems
- Impact on **Data Centers**
  - *"**OpenAI** races to secure more chip capacity: OpenAI announced it'll pay an **unprecedented $30B per year for 4.5 gigawatts of data capacity** from cloud giant Oracle.."*
  - ***"Microsoft locks down $19B in AI cloud capacity"*** (Sept 09th, 2025) [link]
- etc.

Food for Thoughts > [RO4] Agentic Systems Quality

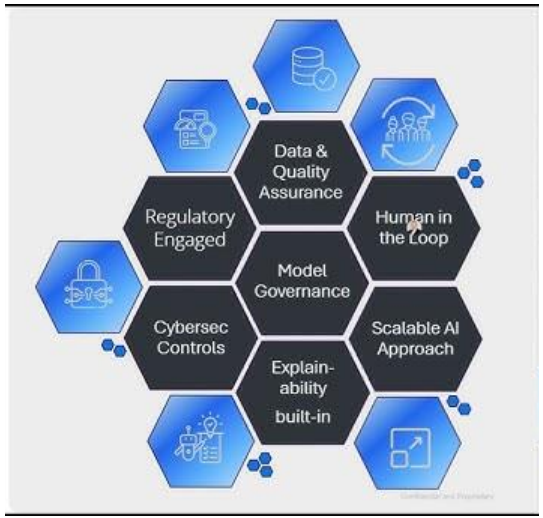


Image from https://www.youtube.com/watch?v=tSZwiiScTF0

How to get assurance on agentic systems quality?

- Reliability and Correctness
  - *LLM-based agents often struggle with consistency and factual accuracy*
- Performance and Efficiency
  - *LLM agents are computationally heavy and can suffer latency or cost issues.*
- Modifiability and Adaptability
  - *Agentic LLM systems pose new challenges for maintainability and evolution*
- Interpretability and Traceability
  - *Understanding the reasoning and decisions of an LLM agent is notoriously difficult*

Food for Thoughts > [RO5] Agentic Systems Engineering in Practice

Business Impact of LLM Agents

How are LLM agents being used in practice?

- Industry Adoption
- Best practices
- Challenges
- Research opportunities
- etc.

Our Ongoing work

- [RO1] Skills and Competencies
- [RO2] Engineering Large-scale MAS
- [RO3] Energy Consumption
- [RO4] Agentic Systems Quality
- [RO5] Agentic Systems Engineering in Practice

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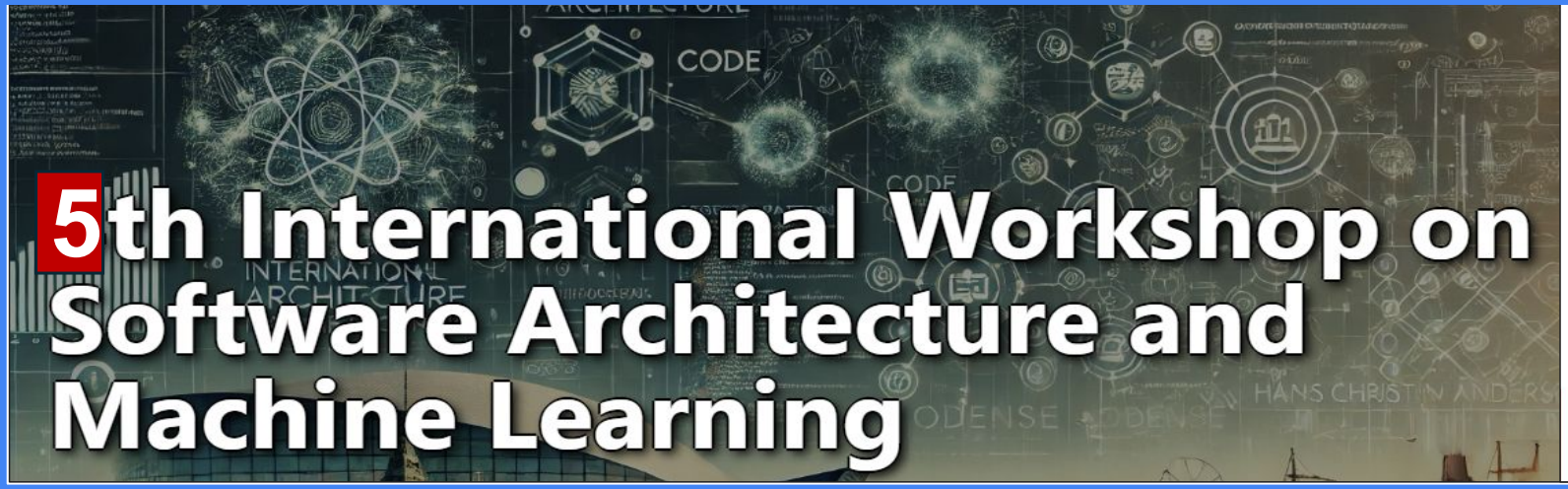
INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY HYDERABAD

Coming next…



ICSA26
Netherlands · Amsterdam
JUNE 22 - 26

CAIN 2026

5th International Conference on AI Engineering
Software Engineering for AI

April 12-13th, 2026
Rio de Janeiro, Brazil

Submission deadlines:
Research Track: October 16/23, 2025
DS, posters, J1C2: November 18, 2025

co-located with ICSE 2026

IEEE Software

Submissions deadline: December 15, 2025

Call for Papers: Special Issue on Engineering Agentic Systems

ACM Transactions on Software Engineering and Methodology

Search within TOSEM

Submissions deadline: November 1, 2025

Special issue on Agentic Artificial Intelligence (AI) in Software

Key Takeaways (for LLM agents)

- LLM agents will provide more **research opportunities** for software engineering research

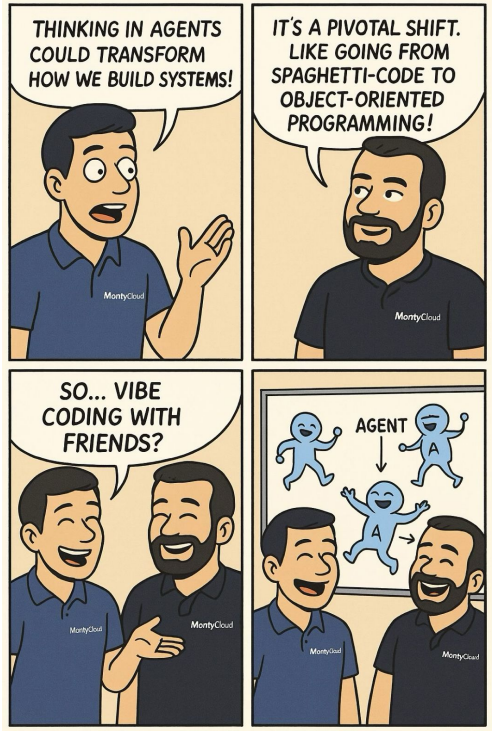
- Let's have an **impact**!
  - It requires understanding what the **industry really needs**!!
    - See the ICSE 2026 track on «Industry Challenges»

- Main RQs:
  - How to engineer **large-scale** agentic systems capable of consistently meeting critical **quality requirements**?
  - How to **migrate** existing software systems to Agentic Systems?

Conclusions and Thoughts

Agentic AI is shifting the way we think about architecting software

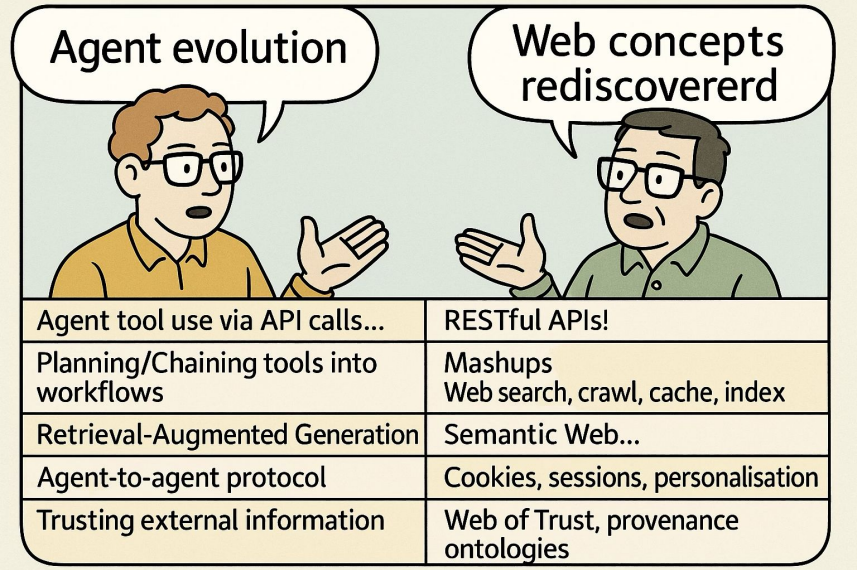
- Exciting times for research in **SA and ML**
- **LLMs are powerful models**
  - Support architectural tasks
  - Generate code, documentation, ADR, etc
- **Agentic AI is about ecosystem**
  - LLM + Memory + tools
  - SE community needs to contribute
  - SA practices needs to be thought about



Credits: Kannan Parthasarathy, MontyCloud Inc.

- Join us @ SAML 2026 **@ICSA 2026**
- Possible working session **@ICSA 2026**

Thank You!



Concept credit: Liming Zhu

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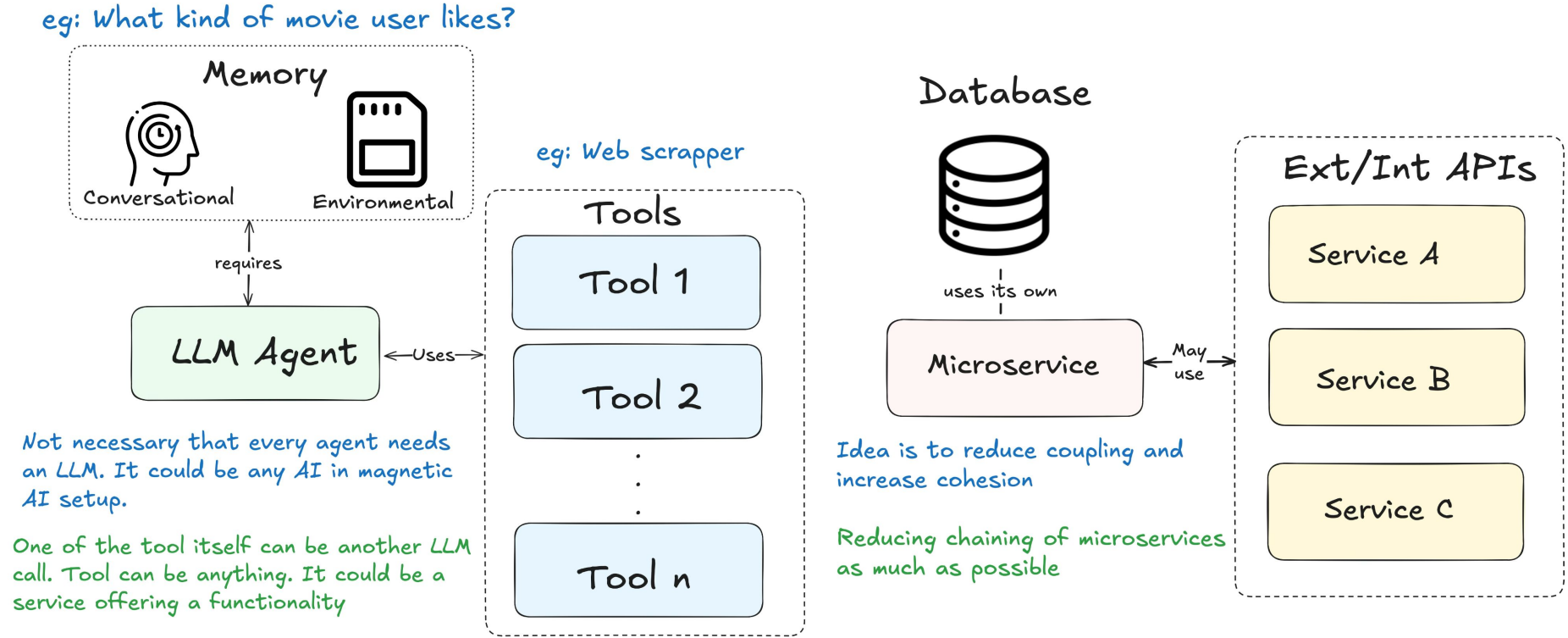
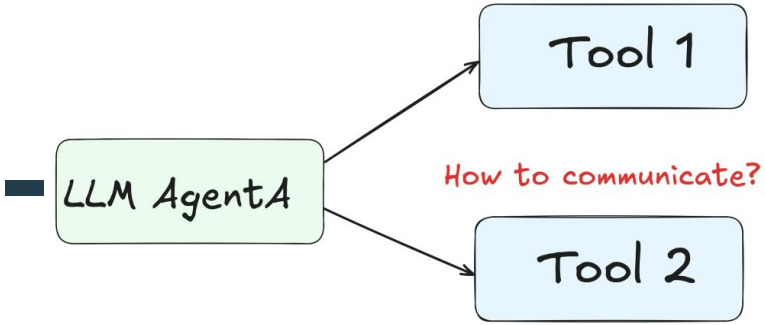
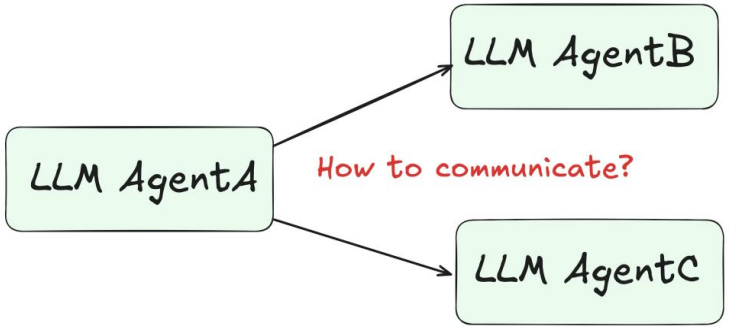
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Agents: What are they made up of

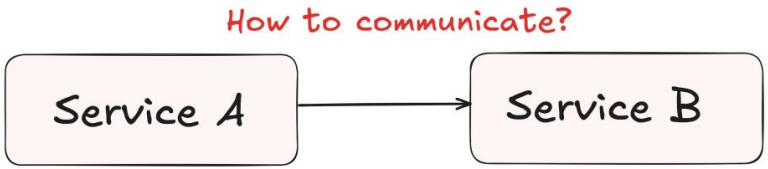




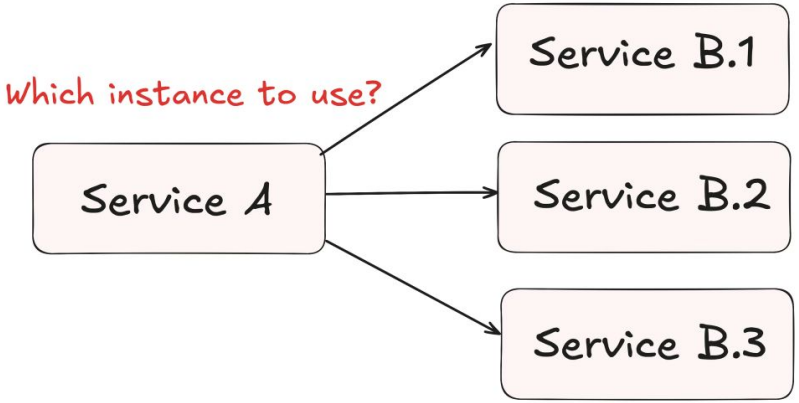
1. Tools can be added dynamically
2. Each tool may have different way of invocation
3. Use protocols like MCP (Model Context Protocol)



1. An agent can be added dynamically
2. Each agent may have different way of invocation
3. Use protocols like A2A (Agent to Agent)



1. Use lightweight protocols like HTTP. API defined
2. Sync vs Async, Orchestration Vs Choreography
3. JSON/Protobuf as the data format



1. Service discovery - Client vs server
2. Services register to Service Registry
3. Eg: Netflix Eureka, Amazon ELB, zookeeper

Landscape of Agentic AI Tools

General Purpose Agentic Frameworks

SWE Agents

Plan for Hands on with Agentic AI

Discussions